
Consistent Learning-to-Defer with Expert-Conditional Advice

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Abstract

1 Learning-to-Defer routes each input to the expert that minimizes expected cost,
2 but it assumes the information available to every expert is fixed at decision time.
3 Modern systems violate this assumption: after selecting an expert, one may also
4 choose what information that expert receives—retrieved documents, tool outputs,
5 or escalation context—and the value of such advice is itself expert-dependent.
6 We introduce *Learning-to-Defer with advice*, a formulation in which the system
7 jointly selects an expert and an expert-conditional advice action. The natural
8 sequential surrogate—one router score and one query head per expert—turns out
9 to be inconsistent: we prove that a broad family of such separated router/query
10 surrogates fails to recover the Bayes rule even in the smallest non-trivial case.
11 We resolve this by an augmented surrogate that operates on the composite expert-
12 advice action space, for which we establish an $\mathcal{H}$ -consistency bound. Across
13 language, tabular, multi-modal, and synthetic benchmarks, the resulting method
14 improves over Learning-to-Defer while adapting its advice-acquisition behavior
15 to the cost regime, and a theorem-aligned synthetic benchmark instantiates the
16 predicted failure mode of separated surrogates.

17 1 Introduction

18 Modern decision systems face two intertwined choices: *who* should handle a given instance, and
19 *what information* that expert should receive. A retrieval-augmented model may answer only after
20 receiving retrieved documents (Guu et al., 2020; Lewis et al., 2020). A tool-augmented agent may
21 be granted access to a calculator, a search engine, or a code interpreter (Schick et al., 2023). These
22 examples concern the second choice; standard Learning-to-Defer (Madras et al., 2018; Mozannar and
23 Sontag, 2020; Mao et al., 2023a; Montreuil et al., 2026b), by contrast, addresses the first. Our setting
24 combines both: the learner must commit to an executed expert–advice pair, where the value of advice
25 depends on the expert who will consume it and the best expert in turn depends on the advice received.

26 Learning-to-Defer (L2D) treats the information available to every expert as fixed at decision time.
27 The entire question of whether and what to acquire is outside the model. This misses an important
28 class of systems in which the value of extra information depends on the expert who will consume it,
29 and can even change which expert is optimal. Figure 3 summarizes this standard L2D pipeline, while
30 Figure 2 shows the deferral-advice protocol considered here. While recent work studies a related
31 binary ask/not-ask problem with a single enriched predictor (Pugnana et al., 2025), our focus is a
32 routed expert–advice decision problem in which advice is expert-conditioned and may come from
33 multiple sources. We call this setting *Learning-to-Defer with advice*: the system jointly selects an
34 expert and decides what information that expert should receive. The central statistical question is
35 then Bayes consistency (Zhang, 2002; Steinwart, 2007; Bartlett et al., 2006): can one learn a policy
36 that recovers the Bayes-optimal expert–advice decision, that is, the expert whose best advice yields
37 the smallest conditional expected cost?

Because routing and advice are coupled — the best advice depends on the expert, and the best expert depends on the advice — one might keep the sequential structure and learn them with separate heads. We show that this natural decomposition is, in general, inconsistent: even with two experts and binary advice, the separated surrogate fails to recover the Bayes rule. The key insight is that the protocol evaluates expert–advice *pairs*, so the surrogate must operate on the same composite action space. Building on this observation, we introduce an augmented surrogate over the joint expert–advice space and prove an $\mathcal{H}$ -consistency guarantee: its minimization is provably aligned with the Bayes-optimal expert–advice decision.

Contributions.

- We introduce *Learning-to-Defer with advice*, a formulation in which the system jointly selects an expert and an expert-conditional advice action, and derive the Bayes-optimal policy, which compares experts at their best-advised cost (Section 4.3, Lemma 3).
- We prove that a broad family of natural separated router-query surrogates is not Bayes consistent, even in the simplest binary-advice setting (Section 4.4, Theorem 6).
- We introduce an augmented surrogate and prove an $\mathcal{H}$ -consistency guarantee together with an excess-risk transfer bound, establishing that surrogate minimization recovers the Bayes-optimal deferral-advice risk in the limit (Section 4.5, Theorem 8, Corollary 9).
- We validate the framework on tabular, language, multi-modal, and synthetic experiments, where it improves over standard Learning-to-Defer, adapts its advice-acquisition rate to the cost regime, and empirically exhibits the failure mode predicted for separated surrogates (Section 5, Appendix C.2).

2 Related Work

L2D extends selective prediction and learning with abstention (Chow, 1970; Bartlett and Wegkamp, 2008; Cortes et al., 2016; Geifman and El-Yaniv, 2017; Cao et al., 2022; Ghosh et al., 2017; Mao et al., 2024a,i; Cortes et al., 2024b) by allowing a learner to defer uncertain inputs to external experts (Madras et al., 2018; Mozannar and Sontag, 2020; Verma et al., 2022; Keswani et al., 2021; Kerrigan et al., 2021; Hemmer et al., 2022; Benz and Rodriguez, 2022; Joshi et al., 2021; Raman and Yee, 2021). A substantial line of work develops surrogate losses and statistical guarantees for this problem (Mozannar and Sontag, 2020; Verma et al., 2022; Cao et al., 2024; Mozannar et al., 2023; Liu et al., 2024; Mao et al., 2024f, 2025d; Charusaie et al., 2022; Mao et al., 2024e; Wei et al., 2024; Montreuil et al., 2026b; Mao, 2025; Montreuil et al., 2026a). The framework has also been extended to regression and multi-task settings and deployed in real systems (Mao et al., 2024g; Strong et al., 2024; Palomba et al., 2025; Montreuil et al., 2025b; Strong et al., 2025; Montreuil et al., 2026d), as well as to robust (Montreuil et al., 2025a, 2026e), online (Montreuil et al., 2026f,c), budgeted or missing experts (DeSalvo et al., 2025; Nguyen et al., 2025), and populations of experts unseen at training time (Tailor et al., 2024; Strong et al., 2026) settings. Our consistency analysis builds on a broader line of work on hypothesis-set-specific consistency, initiated by Long and Servedio (2013); Zhang and Agarwal (2020) and later made quantitative through the $\mathcal{H}$ -consistency bounds of Awasthi et al. (2022), with subsequent refinements and extensions to a wide range of losses (Mao et al., 2024b, 2023b, 2024h, 2025b, 2024c,d; Cortes et al., 2025a,b; Mao et al., 2025c,a; Zhong, 2025).

Recent work has also considered querying human feedback and feeding it to an enriched predictor rather than treating the human only as a deferred decision maker. In the binary ask/not-ask formulation of Pugnana et al. (2025), the learner chooses between one standard predictor and one enriched predictor; there is no routing over multiple experts and no expert-conditional advice policy. They prove realizable consistency for a joint LtA surrogate, while Bayes consistency is left open. We move from this binary ask/not-ask decision to a richer executed-pair decision space, where the learner selects both an expert and an expert-conditional advice action from multiple experts and advice sources. In this setting, we show that natural separated router-query surrogates can fail, and we establish both $\mathcal{H}$ -consistency and Bayes consistency for an augmented surrogate over the executed expert–advice pairs.

3 Preliminaries

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space supporting a pair of random variables (X, Y) with joint distribution $\mathcal{D}$ on $\mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the input space and $\mathcal{Y}$ is an arbitrary measurable output space. We denote by (x, y) a realization of (X, Y) . Throughout, conditional expectations given $X = x$

denote fixed measurable versions, and all finite argmax and argmin operations use smallest-index tie-breaking.

Learning-to-Defer. We consider a fixed collection of J experts, indexed by $[J] := \{1, \dots, J\}$. Each expert $j \in [J]$ is associated with a measurable cost function $c_j : \mathcal{X} \times \mathcal{Y} \rightarrow [0, C]$, that quantifies the expense of routing (x, y) to expert j . This formulation is agnostic to the nature of the expert: an expert may be a predictive model, a language model system, a human annotator, or any other decision mechanism that induces measurable costs. We model expert prediction as a measurable map $e_j : \mathcal{X} \rightarrow \mathcal{Y}$ and decompose the cost as $c_j(x, y) := \psi(e_j(x), y) + \beta_j$, where $\psi : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$ is a measurable task-specific loss (e.g., the 0–1 loss for classification) and $\beta_j \geq 0$ is an expert consultation cost (Madras et al., 2018; Mozannar and Sontag, 2020; Verma et al., 2022). We assume the decomposed costs are uniformly bounded by the same finite constant C above. We denote the tuple of expert predictors by $\mathbf{e} := (e_1, \dots, e_J)$.

Given a collection of fixed, pre-trained experts, L2D seeks a routing policy that assigns each instance to the most cost-effective expert (Madras et al., 2018; Mozannar and Sontag, 2020; Narasimhan et al., 2022; Mao et al., 2023a, 2024g; Montreuil et al., 2025b, 2026b). Let $\mathcal{H}_r \subseteq \{s_r : \mathcal{X} \times [J] \rightarrow \mathbb{R}\}$ denote a hypothesis class of measurable scoring functions for routing. Each $s_r \in \mathcal{H}_r$ induces a measurable router $r : \mathcal{X} \rightarrow [J]$ via $r(x) = \arg \max_{j \in [J]} s_r(x, j)$; we write $\mathbf{s}_r(x) = (s_r(x, j))_{j \in [J]}$ for the full score vector. The deferral loss incurred by routing a realization (x, y) through r is

$$\ell_{\text{def}}(r; x, y, \mathbf{e}) := \sum_{j \in [J]} c_j(x, y) \mathbf{1}\{r(x) = j\}, \quad (1)$$

and the population risk is its expectation $\mathcal{E}_{\ell_{\text{def}}}(r) := \mathbb{E}_{(X, Y) \sim \mathcal{D}}[\ell_{\text{def}}(r; X, Y, \mathbf{e})]$.

$\mathcal{H}_r$ -consistency. The indicator functions in ℓ_{def} make it discontinuous and non-differentiable, precluding direct gradient-based optimization (Zhang, 2002; Steinwart, 2007; Mozannar and Sontag, 2020). Standard practice replaces it with a convex surrogate Φ_{def} that operates on the scoring function s_r and is amenable to gradient-based methods. The surrogate deferral risk is $\mathcal{E}_{\Phi_{\text{def}}}(s_r) = \mathbb{E}_{(X, Y) \sim \mathcal{D}}[\Phi_{\text{def}}(s_r; X, Y, \mathbf{e})]$, with best-in-class value $\mathcal{E}_{\Phi_{\text{def}}}^*(\mathcal{H}_r) := \inf_{s_r \in \mathcal{H}_r} \mathcal{E}_{\Phi_{\text{def}}}(s_r)$, and let $\mathcal{E}_{\ell_{\text{def}}}^B(\mathcal{H}_r) := \inf_{s_r \in \mathcal{H}_r} \mathcal{E}_{\ell_{\text{def}}}(r_{s_r})$ denote the best achievable true risk within $\mathcal{H}_r$, where r_{s_r} is the router induced by s_r . To obtain population-level excess-risk control, Awasthi et al. (2022) introduced $\mathcal{H}$ -consistency bounds, subsequently refined and extended in (Mao et al., 2024b,h, 2025b,a).

Definition 1 ($\mathcal{H}_r$ -consistency bound). *The surrogate Φ_{def} satisfies an $\mathcal{H}_r$ -consistency bound with respect to ℓ_{def} if there exists a non-decreasing function $\Gamma : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $\Gamma(0) = 0$ such that for all $s_r \in \mathcal{H}_r$,*

$$\mathcal{E}_{\ell_{\text{def}}}(r) - \mathcal{E}_{\ell_{\text{def}}}^B(\mathcal{H}_r) + \mathcal{U}_{\ell_{\text{def}}}(\mathcal{H}_r) \leq \Gamma(\mathcal{E}_{\Phi_{\text{def}}}(s_r) - \mathcal{E}_{\Phi_{\text{def}}}^*(\mathcal{H}_r) + \mathcal{U}_{\Phi_{\text{def}}}(\mathcal{H}_r)), \quad (2)$$

where r is the router induced by s_r . For a surrogate loss L depending on s_r , or a true loss evaluated at the router induced by s_r , the associated minimizability gap is $\mathcal{U}_L(\mathcal{H}_r) := \inf_{s_r \in \mathcal{H}_r} \mathcal{E}_L(s_r) - \mathbb{E}_X[\inf_{s_r \in \mathcal{H}_r} \mathbb{E}_{Y|X}[L(s_r; X, Y, \mathbf{e})]]$, applied here to $L \in \{\ell_{\text{def}}, \Phi_{\text{def}}\}$.

When $\mathcal{H}_r$ is rich enough for these gaps to vanish, the bound recovers Bayes consistency (Steinwart, 2007; Awasthi et al., 2022). In the standard L2D setting, such bounds have been established for the comp-sum family of cross-entropy surrogates (Mao et al., 2024e,g, 2023a; Montreuil et al., 2025b, 2026b), guaranteeing that surrogate minimization over a rich enough class recovers the Bayes-optimal router.

4 Problem Formulation and Surrogate Design

The deferral framework of Section 3 assumes that every expert acts on the input x alone. We now extend this setting by allowing the system to acquire additional information—advice—and provide it to the chosen expert before that expert acts. This introduces a genuine statistical complication: the policy must choose both the expert and the advice source from x alone, while the advice is revealed only after that commitment. We therefore first formalize the protocol and its Bayes rule, and then study which surrogate constructions remain valid in this extended setting.

4.1 Advice Sources

Advice acquisition introduces two coupled difficulties. First, the value of a source is expert-dependent: the same retrieved passage or tool call may help one expert and be irrelevant, or even harmful, for

another (Guu et al., 2020; Lewis et al., 2020; Schick et al., 2023). The choice of *what* to acquire therefore cannot be assessed independently of the choice of *who* receives it. Second, both decisions must be made from x alone. The realized advice value is revealed only after the expert and advice source have already been selected. The difficulty is therefore not merely that the protocol is sequential, but that learning must target a hidden executed expert-advice pair rather than two independently observable decisions.

Formal setup. We formalize this uncertainty by augmenting the data-generating process with K advice variables. The triple (X, A, Y) is jointly distributed, where $A = (A^1, \dots, A^K)$ and each A^k takes values in a measurable space $\mathcal{A}_k$. Let $\mathcal{A} := \mathcal{A}_1 \times \dots \times \mathcal{A}_K$, and write (x, a, y) for a realization. This model accommodates both deterministic advice, such as a database lookup, and stochastic advice, such as a language model output (Touvron et al., 2023; OpenAI et al., 2024). The action $k = 0$ denotes the decision to acquire no advice, and $[K]_0 := \{0, 1, \dots, K\}$ is the resulting advice-action set.

Masking. The protocol enforces that at most one source is revealed per instance. Fix a missing-advice symbol $\perp \notin \bigcup_{k=1}^K \mathcal{A}_k$ and define $\tilde{\mathcal{A}} := (\mathcal{A}_1 \cup \{\perp\}) \times \dots \times (\mathcal{A}_K \cup \{\perp\})$. For each $k \in [K]_0$, the masking operator $m_k : \mathcal{A} \rightarrow \tilde{\mathcal{A}}$ retains the k -th coordinate and replaces all others with $\perp$; when $k = 0$ every coordinate is masked. We write $\tilde{a}^{(k)} := m_k(a)$ and $\tilde{A}^{(k)} := m_k(A)$ for the masked advice.

4.2 The Deferral-Advice Loss

With advice sources and masking in place, we now define how the system acts and what it pays. The goal is to formalize the true loss of the protocol and identify the Bayes-optimal policy.

Costs. In the standard setting, routing to expert j incurs a cost $c_j(x, y)$ that captures both prediction error and consultation fee. With advice, the cost acquires a new dimension: the system may spend an acquisition fee γ_k to provide advice to the expert, hoping that the additional information reduces the expert’s prediction error by more than it costs to obtain. For each pair $(j, k) \in [J] \times [K]_0$, let $c_{j,k} : \mathcal{X} \times \mathcal{A} \times \mathcal{Y} \rightarrow [0, C]$ be a measurable cost function, uniformly bounded by $C < \infty$.

$$c_{j,k}(x, a, y) := \underbrace{\psi(e_j(x, \tilde{a}^{(k)}), y)}_{\text{task loss}} + \underbrace{\beta_j}_{\text{expert fee}} + \underbrace{\gamma_k}_{\text{advice fee}}, \quad (3)$$

where $e_j : \mathcal{X} \times \tilde{\mathcal{A}} \rightarrow \mathcal{Y}$ is measurable and extends the expert prediction of Section 3 to accept masked advice $\tilde{a}^{(k)}$ —reducing to $e_j(x)$ when $k = 0$ —and $\beta_j \geq 0$ is the expert consultation cost, while $\gamma_k \geq 0$ is the advice acquisition cost with $\gamma_0 = 0$. The no-advice compatibility condition $c_{j,0}(x, a, y) = c_j(x, y)$ holds by construction. The trade-off is explicit: acquiring advice k incurs the fee γ_k but may reduce the task loss ψ if the advice is informative for expert j . Whether the trade-off is favorable depends jointly on the expert, the advice source, and the input.

Policy and true loss. A deferral-advice policy is a pair of measurable maps $r : \mathcal{X} \rightarrow [J]$ and $q : \mathcal{X} \times [J] \rightarrow [K]_0$. Let $\mathcal{H}_q \subseteq \{s_q : \mathcal{X} \times [J] \times [K]_0 \rightarrow \mathbb{R}\}$ be a hypothesis class of measurable query scoring functions; each $s_q \in \mathcal{H}_q$ induces a query function via $q(x, j) = \arg \max_{k \in [K]_0} s_q(x, j, k)$. We write $\mathbf{S}_q(x) \in \mathbb{R}^{J \times (K+1)}$ for the matrix with entries $s_q(x, j, k)$, so that each row corresponds to one expert and the row-wise argmax gives the advice index for that expert. Upon observing x , the system routes to expert $j = r(x)$ and selects advice index $k = q(x, j)$; the masked advice $\tilde{A}^{(k)}$ is then revealed to expert j . Although $q(x, \cdot)$ is defined for every expert, only the executed value $q(x, r(x))$ enters the realized loss (see Algorithm 2 and Figure 2; Figure 3 contrasts this with the usual L2D pipeline).

Definition 2 (True deferral-advice loss). *Given a policy (r, q) , the true deferral-advice loss on a realization (x, a, y) is*

$$\ell_{\text{def-adv}}(r, q; x, a, y, \mathbf{e}) := \sum_{j \in [J]} \sum_{k \in [K]_0} c_{j,k}(x, a, y) \mathbf{1}\{r(x) = j\} \mathbf{1}\{q(x, j) = k\}.$$

The double indicator selects exactly one expert-advice pair per instance. The loss generalizes (1) from J to $J(K+1)$ actions and depends on the policy only through the executed pair $(r(x), q(x, r(x)))$. Since $c_{j,0}(x, a, y) = c_j(x, y)$ by construction, setting $K = 0$ recovers the standard deferral loss ℓ_{def} (Mozannar and Sontag, 2020; Mao et al., 2023a; Montreuil et al., 2026b). A concrete numerical example illustrating the loss is given in Example 1.

4.3 Deferral with Advice Dominates Standard Deferral

We now derive the Bayes-optimal policy. Its structure reveals two properties: the choice of advice is inherently coupled to the choice of expert, and deferral with advice is more powerful than standard Learning-to-Defer.

The Bayes-optimal policy. In Section 4.1 we argued that the value of advice is expert-dependent, and the policy of Definition 2 reflects this by conditioning the query $q(x, j)$ on the expert. We now show that minimizing the expected loss of Definition 2 recovers the desired policy. To determine which expert is best, the system must first evaluate each expert at its best-advised cost—this requires solving the advice selection problem for every expert independently. The optimal query *necessarily* depends on the expert, since the best advice for expert j is generally different from the best advice for expert j' . The router then compares experts on their optimally-advised costs rather than their unaided ones.

Lemma 3 (Bayes-optimal deferral-advice policy). *There exists a measurable Bayes-optimal policy (r^*, q^*) that minimizes $\mathbb{E}[\ell_{\text{def-adv}}(r, q; X, A, Y, \mathbf{e})]$ over all measurable policies and, for $\mathbb{P}_X$ -a.e. x , can be chosen by smallest-index tie-breaking as follows:*

1. The Bayes query selects the best advice for each expert:

$$q^*(x, j) = \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x]. \quad (4)$$

2. The Bayes router selects the best optimally-advised expert:

$$r^*(x) = \min_{j \in [J]} \mathbb{E}[c_{j,q^*(x,j)}(X, A, Y) \mid X = x]. \quad (5)$$

The proof is given in Appendix B.1. The result reveals a fundamental coupling: the router does not compare experts on their unaided costs $\mathbb{E}[c_j(X, Y) \mid X = x]$ as in the standard setting, but on their optimally-advised ones. An expert that would never be selected under standard deferral may become optimal once paired with the right advice.

When is advice worth acquiring? In any practical system, acquiring information has a cost, and a sensible policy should query only when the expected benefit justifies the expense. The smallest-index Bayes query of Lemma 3 satisfies exactly this principle. Under the cost decomposition (3), the system acquires advice $k \geq 1$ for expert j if and only if some queried source strictly beats the no-advice option in expected total cost.

Lemma 4 (Advice acquisition condition). *Under the cost decomposition (3) and the smallest-index tie-breaking of Lemma 3, the Bayes-optimal query satisfies $q^*(x, j) \neq 0$ if and only if there exists $k \in [K]$ such that*

$$\mathbb{E}[\psi(e_j(X, \tilde{A}^{(0)}), Y) \mid X = x] - \mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X = x] > \gamma_k. \quad (6)$$

When no source satisfies this condition, $q^(x, j) = 0$ and the expert acts on x alone.*

The proof is given in Appendix B.2. The left-hand side of (6) measures how much advice k improves expert j 's prediction; the right-hand side is the price of obtaining it. The system acquires advice only when it is expected to pay for itself.

Deferral with advice dominates L2D. Since the no-advice action $k=0$ is always available, the learner can recover the standard L2D decision whenever advice is not worth its cost. The point is stronger: for each expert the best advised cost is at most the unaided cost; propagating this through the comparison over experts yields a strict population-level improvement whenever advice lowers the best achievable conditional cost on a set of positive measure.

Lemma 5 (Deferral with advice dominates standard deferral). *If $\mathcal{E}_{\text{def}}^*$ denotes the Bayes risk of the standard deferral setting of Section 3, then*

$$\mathcal{E}_{\text{def-adv}}^* \leq \mathcal{E}_{\text{def}}^*. \quad (7)$$

The proof, including a necessary and sufficient condition for strict inequality, is given in Appendix B.3.

4.4 The Separated Surrogate Fails

Because the protocol is sequential, the natural first attempt is to learn it sequentially as well: one router score, one query head per expert, and smooth logistic replacements for the hard indicators (Bartlett et al., 2006). This is precisely the construction a learning-theory practitioner would derive by applying the standard recipe—replacing each indicator in the true loss by a smooth convex surrogate term—to the product of routing and query decisions. But does mirroring the control flow suffice for consistency? We show that a broad family of separated surrogates already fails in the smallest non-trivial case: two experts ($J = 2$) and one binary advice ($K = 1$).

Let $s_r^b : \mathcal{X} \rightarrow \mathbb{R}$ be a binary router score and, for each expert $j \in \{1, 2\}$, let $s_{q_j^b} : \mathcal{X} \rightarrow \mathbb{R}$ be a binary query score. Writing $\mathbf{s}_{q^b} = (s_{q_1^b}, s_{q_2^b})$, the induced decisions are $r^b(x) = 1 + \mathbf{1}\{s_r^b(x) \geq 0\}$ and $q_j^b(x) = \mathbf{1}\{s_{q_j^b}(x) \geq 0\}$.

The natural separated construction. In standard learning theory, one often replaces hard indicators by smooth convex surrogates (Zhang, 2002; Bartlett et al., 2006; Steinwart, 2007). Applying this recipe to the binary executed-pair loss yields logistic potentials for the decoded binary outcomes, $\Phi_0(u) = \log(1 + e^{-u})$ and $\Phi_1(u) = \log(1 + e^u)$. Rather than analyzing only the basic surrogate obtained by directly substituting these terms, we study a strictly larger family with monotone cost transforms, reparameterized query scores, and flexible positive router weights. Let $\nu : [0, C] \rightarrow [0, \infty)$ be a monotone cost transform, $G : \mathbb{R} \rightarrow (0, \infty)$ a query amplitude, and $U : \mathbb{R} \rightarrow \mathbb{R}$ a strictly increasing query reparameterization. The resulting *router-query separated surrogate* family is

$$\Phi_{\text{def-adv}}^{r/q}(\mathbf{s}_r^b, \mathbf{s}_{q^b}) = \sum_{j=1}^2 \sum_{k=0}^1 \Psi_j(s_r^b(x)) \nu(c_{j,k}(x, a, y)) G(s_{q_j^b}(x)) \Phi_k(U(s_{q_j^b}(x))). \quad (8)$$

Setting $\nu = \text{id}$, $G \equiv 1$, $U = \text{id}$, and $\Psi_j = \Phi_{j-1}$ recovers the basic construction obtained by assigning these logistic potentials to the corresponding decoded binary outcomes—the standard recipe in cost-sensitive surrogate design (Cortes et al., 2016, 2024a); we give the full derivation in Appendix B.4.1.

Where the mismatch enters. This separation has one decisive consequence. After profiling over the query heads, each expert is summarized by a scalar quantity

$$F(u, v) := \inf_{t \in \mathbb{R}} [\nu(u) G(t) \Phi_0(U(t)) + \nu(v) G(t) \Phi_1(U(t))] \quad (9)$$

that depends on the *entire* row, not only on its best entry. The Bayes rule, by contrast, compares row minima: $\min\{c_{1,0}, c_{1,1}\}$ versus $\min\{c_{2,0}, c_{2,1}\}$ (cf. (5)). The surrogate and the Bayes rule compress each row in fundamentally different ways. That difference is the entire obstruction.

We formalize this failure at the level of Bayes consistency: every population surrogate minimizer should decode to a Bayes-optimal action (Zhang, 2002; Bartlett et al., 2006; Steinwart, 2007; Tewari and Bartlett, 2007).

Theorem 6 (Broad Bayes inconsistency of separated router/query surrogates). *Under mild regularity conditions on ν , G , U , Ψ_1 , Ψ_2 (Appendix B.4), for every $b \in (0, C)$ and $\varepsilon \in (0, C - b)$, there exists $\delta \in (0, b)$ such that the pointwise conditional cost table*

$$\bar{\mathbf{C}}(x) = \begin{pmatrix} b - \delta & C \\ b & b + \varepsilon \end{pmatrix} \quad (10)$$

has Bayes-optimal expert $j^* = 1$, while the unique minimizer of the pointwise surrogate (8) decodes to expert 2. In particular, the entire family (8) is not Bayes consistent for the true loss.

Expert 1 has one excellent action and one catastrophic one, whereas expert 2 has two mediocre actions. Bayes compares row minima, ignores the catastrophic entry, and therefore prefers expert 1. The separated surrogate behaves differently. After profiling out the query head, row 1 is summarized by $F(b - \delta, C)$, which still depends on the catastrophic entry. Appendix B.4.4 shows that $F(u, v)$ is strictly increasing in v for fixed $u \in (0, C)$, so this distortion is unavoidable. The router therefore compares profiled row summaries rather than row minima and selects expert 2.

The contradiction already appears in the smallest non-trivial setting ($J=2$, $K=1$) and persists across the entire parameterization family (8). Because this family encompasses every modification the

standard surrogate-design recipe suggests, the result rules out the entire class of constructions a practitioner would naturally derive from the product-of-indicators loss. Appendix C.2 complements the theorem with a synthetic experiment where the separated surrogate’s excess risk stays bounded away from zero.

4.5 The Augmented Surrogate

The separated surrogate of Section 4.4 fails because profiling the query heads row by row distorts the router’s comparison. This failure makes surrogate consistency nontrivial: even in the simpler binary ask/not-ask decision structure of Pugnana et al. (2025), realizable consistency is known but Bayes consistency is left open. We show that such guarantees can nevertheless be obtained in our setting by abandoning the separated construction and learning over the executed expert–advice pairs directly.

Composite actions. Recall that the loss depends only on the executed pair $(r(x), q(x, r(x)))$. Define the composite action set $\Pi := [J] \times [K]_0$, of cardinality $|\Pi| = J(K + 1)$. Each action $(j, k) \in \Pi$ represents the joint decision to route to expert j and acquire advice k . A composite predictor is a measurable map $\pi : \mathcal{X} \rightarrow \Pi$. Let $\mathcal{H}_\pi \subseteq \{s_\pi : \mathcal{X} \times \Pi \rightarrow \mathbb{R}\}$ be a hypothesis class of measurable scoring functions; each $s_\pi \in \mathcal{H}_\pi$ induces a composite predictor via $\pi(x) = \arg \max_{(j,k) \in \Pi} s_\pi(x, (j, k))$. We write $\mathbf{s}_\pi(x) = (s_\pi(x, (j, k)))_{(j,k) \in \Pi}$ for the full score vector.

A sequential policy (r, q) induces the composite action $\pi(x) = (r(x), q(x, r(x)))$, and every composite action (j, k) can in turn be implemented by a sequential policy with the same realized loss. The composite view simply makes explicit what the loss has depended on all along: the executed expert–advice pair. Proposition 17, proved in Appendix B.5, shows that the sequential and composite formulations therefore induce the same population risks and share the same Bayes-optimal decisions as those identified in Lemma 3.

The augmented surrogate. The composite view aligns the learning problem with the Bayes comparison itself, but the true loss remains non-differentiable. The deferral–advice loss is not a uniform 0–1 loss over Π : it is cost-sensitive, with instance- and action-dependent weights. The key step (Appendix B.6) decomposes the realized loss into an action-independent offset and a weighted mismatch term:

$$\ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e}) = D(x, a, y) + \sum_{i \in \Pi} w_i(x, a, y) \mathbf{1}\{\pi(x) \neq i\}, \quad (11)$$

where $D(x, a, y)$ is independent of π and $w_i(x, a, y) := \max_{i' \in \Pi} c_{i'}(x, a, y) - c_i(x, a, y)$. Let Φ_{01}^τ denote the comp-sum multiclass surrogate of Mao et al. (2023b), parameterized by $\tau \geq 0$; at $\tau = 1$ it recovers the standard logistic (cross-entropy) loss. Applying Φ_{01}^τ to the weighted mismatch term yields the augmented deferral–advice surrogate, which we derive in Appendix B.7:

Lemma 7 (Augmented surrogate).

$$\Phi_{\text{def-adv}}^{\text{aug}, \tau}(s_\pi; x, a, y, \mathbf{e}) := \sum_{i \in \Pi} w_i(x, a, y) \Phi_{01}^\tau(\mathbf{s}_\pi(x), i). \quad (12)$$

Consistency: the main positive result. Lemma 7 gives the right starting point, but smoothly replacing mismatch indicators is not enough: the separated surrogate of Section 4.4 can still decode to the wrong executed decision (Theorem 6). The surrogate must preserve the comparison made by the protocol.

The augmented surrogate succeeds where the separated one fails because it never profiles out a subset of scores. The separated construction optimizes query scores row by row and hands the router a scalar summary per expert; that summary depends on the non-minimal entry and distorts the comparison (Theorem 6). The augmented surrogate, by contrast, assigns one score to each composite action (j, k) and compares them all at once. No row-by-row reduction ever occurs, so the surrogate comparison operates on the same objects as the Bayes comparison.

Theorem 8 ($\mathcal{H}_\pi$ -consistency of the augmented surrogate). *Assume that $\mathcal{H}_\pi$ satisfies the standard symmetry and completeness conditions required by the comp-sum multiclass $\mathcal{H}_\pi$ -consistency theorem for Φ_{01}^τ on the label set Π . For every $\tau \geq 0$, the augmented surrogate $\Phi_{\text{def-adv}}^{\text{aug}, \tau}$ satisfies an $\mathcal{H}_\pi$ -consistency bound with respect to $\ell_{\text{def-adv}}$. Specifically, for every $s_\pi \in \mathcal{H}_\pi$ and its induced composite predictor π ,*

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\pi) - \mathcal{E}_{\ell_{\text{def-adv}}}^B(\mathcal{H}_\pi) + \mathcal{U}_{\ell_{\text{def-adv}}}(\mathcal{H}_\pi) \leq \tilde{\Gamma}_\tau \left(\mathcal{E}_{\Phi_{\text{def-adv}}^{\text{aug}, \tau}}(s_\pi) - \mathcal{E}_{\Phi_{\text{def-adv}}^{\text{aug}, \tau}}^*(\mathcal{H}_\pi) + \mathcal{U}_{\Phi_{\text{def-adv}}^{\text{aug}, \tau}}(\mathcal{H}_\pi) \right). \quad (13)$$

318 With $\tilde{\Gamma}_\tau(u) := \mathbb{E}[\|\mathbf{w}(X)\|_1] \Gamma_\tau\left(\frac{u}{\mathbb{E}[\|\mathbf{w}(X)\|_1]}\right)$, where Γ_τ is the non-negative, non-decreasing,
 319 concave transfer function for Φ_{01}^τ on Π , and $\mathbf{w}(x) := (\mathbb{E}[w_i(X, A, Y) \mid X = x])_{i \in \Pi}$.

320 We prove this theorem in Appendix B.8. For the standard log-softmax (cross-entropy) surrogate
 321 ($\tau = 1$), $\Gamma_1(u) = \sqrt{2u}$, so the true excess risk decays as $O(\sqrt{\mathbb{E}[\|\mathbf{w}(X)\|_1] \varepsilon})$ in the surrogate
 322 excess risk ε , at the standard calibration rate (Mao et al., 2023b, 2024h). The weights $w_i(x, a, y)$
 323 are both instance- and action-dependent, so the surrogate defines a *weighted* cost-sensitive problem
 324 rather than a uniform one. The technical core is a reweighting argument that absorbs these weights
 325 into an auxiliary distribution $\tilde{\mathcal{D}}$ on $\mathcal{X} \times \Pi$ (Lemma 19 in the appendix). The transfer function $\tilde{\Gamma}_\tau$
 326 that emerges is problem-adaptive: it depends on the expected weight norm $\mathbb{E}[\|\mathbf{w}(X)\|_1]$, which
 327 encodes the cost structure of the particular expert-advice environment. When no advice is available
 328 ($K = 0$, so $\Pi = [J]$), the bound (13) recovers the $\mathcal{H}_r$ -consistency bounds established for standard
 329 Learning-to-Defer by Mao et al. (2023a, 2024g); Montreuil et al. (2025b, 2026b).

330 **Corollary 9** (Bayes consistency). *Assume the hypotheses of Theorem 8. If the minimizability gaps*
 331 *in Theorem 8 vanish, the class is Bayes-rich in the sense that $\mathcal{E}_{\ell_{\text{def-adv}}}^B(\mathcal{H}_\pi) = \inf_\pi \mathcal{E}_{\ell_{\text{def-adv}}}(\pi)$,*
 332 *and $\mathcal{E}_{\Phi_{\text{def-adv}}^{\text{aug}, \tau}}(\hat{s}_{\pi, n}) - \mathcal{E}_{\Phi_{\text{def-adv}}^*}^{\text{aug}, \tau}(\mathcal{H}_\pi) \rightarrow 0$ in probability, then the induced predictors $\hat{\pi}_n$ satisfy*
 333 *$\mathcal{E}_{\ell_{\text{def-adv}}}(\hat{\pi}_n) - \inf_\pi \mathcal{E}_{\ell_{\text{def-adv}}}(\pi) \rightarrow 0$ in probability.*

334 Thus, whenever the hypothesis class is rich enough and the surrogate is well-optimized, the induced
 335 policy converges in risk to the Bayes-optimal deferral-advice risk of Lemma 3. When $\mathcal{H}_\pi$ contains
 336 all measurable functions, both minimizability gaps vanish (Zhang and Agarwal, 2020; Bartlett et al.,
 337 2006; Long and Servedio, 2013) and the bound reduces to classical Bayes-consistency. The separated
 338 surrogate offers no such guarantee.

339 5 Experiments

340 We evaluate the augmented surrogate on four benchmarks spanning language, tabular, multi-modal,
 341 and synthetic domains. (i) **FEVER** (Thorne et al., 2018): fact verification with LLM and NLI
 342 experts receiving retrieved evidence (detailed below). (ii) **Sensitive escalation** (Howard et al., 2019):
 343 fraud detection where advice reveals progressively more sensitive feature groups (Appendix C.4).
 344 (iii) **CLIP prompt escalation** (Radford et al., 2021): zero-shot image classification where advice
 345 enlarges the prompt family (Appendix C.5). (iv) **Synthetic**: a theorem-aligned benchmark that
 346 isolates the failure mode of separated surrogates predicted by Theorem 6 (Appendix C.2).

347 5.1 FEVER setup

348 **Task, experts, and advice.** We use the FEVER fact-verification benchmark (Thorne et al., 2018).
 349 The expert pool contains Qwen3-4B-Instruct, Qwen3-8B (Yang et al., 2025), and DeBERTa-v3-
 350 large (He et al., 2021), with consultation costs $\beta = (0.03, 0.05, 0.04)$. Advice consists of retrieved
 351 evidence. The five actions are no retrieval, BM25 top-1, top-3, top-5, and a query-reformulation
 352 pipeline produced by Qwen2.5-1.5B. The base advice costs are $\gamma^{\text{base}} = (0, 0.015, 0.02, 0.03, 0.01)$,
 353 and the deployed cost is $\gamma_k = \lambda \gamma_k^{\text{base}}$ for $\lambda \in \{0, 0.5, 1, 2, 4, 8, 10\}$.

354 **Training protocol and baselines.** We evaluate the empirical validation average of $\mathcal{E}_{\ell_{\text{def-adv}}}(\pi)$. We
 355 compare against four baselines, each isolating a different aspect of the decision: (i) standard L2D
 356 (Mao et al., 2023a), restricted to $k=0$, which can route but never acquire advice; (ii) the best fixed
 357 expert-advice pair, an oracle over non-adaptive policies; (iii) two partial-randomization ablations
 358 — learned expert with random advice, and learned advice with random expert — that separate the
 359 contribution of each decision component; and (iv) a fully random (j, k) baseline as an uninformed
 360 floor.

361 5.2 Results

362 The value of advice is strongly expert-dependent: on this three-class task, without retrieval all three
 363 experts hover around 33% accuracy, but their preferred retrieval actions diverge—DeBERTa is best
 364 with top-1, whereas both LLMs prefer top-5 (full per-expert table in Appendix C.3).

365 Table 1 gives the main result. Our method achieves the lowest mean true loss in seven of the eight
 366 cost regimes, with the largest gain at $\lambda = 0$ (0.064 over L2D); as advice cost grows the policy shifts
 367 toward no-query decisions, and at $\lambda = 10$ it recovers the L2D baseline. It also stays below the best

Table 1: Validation true deferral-advice loss across advice-cost multipliers λ . Entries are mean $\pm$ std over 4 seeds; lower is better. Bold marks the best mean per row.

λ	Ours	L2D	Best fixed	Learned j , rand. k	Rand. j , learned k	Rand. (j, k)
0	0.555 $\pm$ 0.002	0.619 $\pm$ 0.001	0.611	0.633 $\pm$ 0.005	0.640 $\pm$ 0.007	0.675 $\pm$ 0.006
0.5	0.558 $\pm$ 0.002	0.619 $\pm$ 0.001	0.619	0.640 $\pm$ 0.005	0.657 $\pm$ 0.007	0.682 $\pm$ 0.006
1	0.566 $\pm$ 0.001	0.619 $\pm$ 0.001	0.626	0.647 $\pm$ 0.007	0.664 $\pm$ 0.007	0.690 $\pm$ 0.006
2	0.580 $\pm$ 0.001	0.619 $\pm$ 0.001	0.641	0.662 $\pm$ 0.006	0.674 $\pm$ 0.009	0.705 $\pm$ 0.006
4	0.596 $\pm$ 0.003	0.619 $\pm$ 0.001	0.671	0.695 $\pm$ 0.006	0.695 $\pm$ 0.007	0.735 $\pm$ 0.007
8	0.613 $\pm$ 0.001	0.619 $\pm$ 0.001	0.695	0.760 $\pm$ 0.006	0.711 $\pm$ 0.006	0.795 $\pm$ 0.007
10	0.619 $\pm$ 0.001	0.619 $\pm$ 0.001	0.695	0.788 $\pm$ 0.006	0.715 $\pm$ 0.008	0.825 $\pm$ 0.007

fixed expert–advice pair throughout, confirming that the gain stems from input-dependent selection rather than from mere access to a larger action set. The partial-randomization ablations further show that both decision halves matter, and that *Random j , learned k* beats fully random — evidence that the surrogate learns a genuinely expert-conditional advice strategy rather than a global retrieval preference.

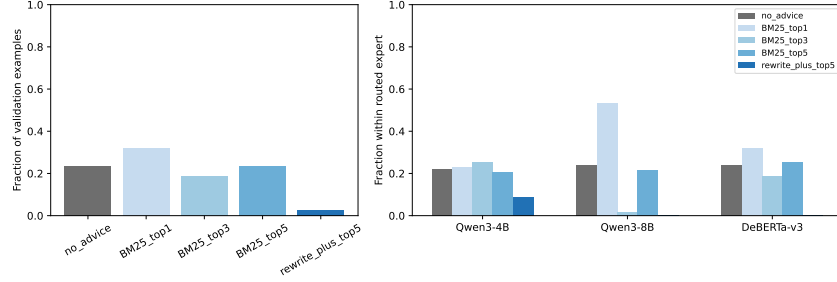


Figure 1: Deployed advice distribution of our method at $\lambda = 0$ in FEVER. The left panel shows the overall fraction of validation examples assigned to each advice action. The right panel conditions on the routed expert.

5.3 Additional benchmarks

Sensitive escalation (tabular). On the fraud-detection benchmark (Howard et al., 2019), advice reveals progressively more sensitive feature groups to one of three experts (logistic regression, GBDT, MLP). Our method achieves the lowest true loss at every cost multiplier (Table 11 in Appendix C.4), improving over L2D by 0.022 at $\lambda=0$ and converging to it at $\lambda=5$. The best fixed pair itself changes with the cost regime, confirming that the optimal policy is inherently cost-dependent.

CLIP prompt escalation (multi-modal). On ImageNet-1k (Russakovsky et al., 2015), advice enlarges the prompt family used by frozen CLIP experts (B/32, B/16, L/14) from a single template to ensembles of up to 32. Our method attains the lowest true loss across the entire low- and moderate-cost regime (Table 16 in Appendix C.5), reducing L2D’s 0.352 to 0.337 at $\lambda=0$ and matching L2D at $\lambda=10$. The three experts prefer different prompt families at different cost levels, precisely the regime where a joint expert-advice policy is most natural.

Synthetic (theorem-aligned). A binary-setting benchmark ($J=2$, $K=1$) instantiates the cost tables from Theorem 6. At $n=5000$, our method reaches true risk 0.281, essentially matching the Bayes optimum 0.280, while the separated surrogate stalls at 0.334 and L2D at 0.342 (Table 4 in Appendix C.2). Regionwise, the separated surrogate misroutes on the theorem region (18% Bayes-action match), L2D cannot realize the queried Bayes action on the advice-helpful region (0%), and our method achieves 99.3% on both.

6 Conclusion

We introduced *Learning-to-Defer with advice*, a setting in which the system must decide not only which expert should act, but also which additional information that expert should receive. A broad family of natural separated router-query surrogates is not Bayes consistent even in the smallest non-trivial case; we gave an augmented surrogate that learns directly over the composite expert-advice action space and proved an $\mathcal{H}$ -consistency guarantee. Empirically, the resulting method improves over Learning-to-Defer across language, tabular, and multi-modal benchmarks while adapting its advice acquisition to the cost regime. Once information acquisition is expert dependent, Learning-to-Defer must reason not only about *who* acts, but also about *what they get to see*.

7 Limitations

Training requires precomputed costs $c_{j,k}(x, a, y)$ for every expert–advice pair. This full-information assumption is standard in Learning-to-Defer (Mozannar and Sontag, 2020; Mao et al., 2023a; Verma et al., 2022; Montreuil et al., 2026b), where it takes the form of precomputed expert predictions on the training set; at inference time, only the single selected expert–advice pair is executed. As the first formulation of Learning-to-Defer with advice and the first consistency guarantee for its composite action space, the present work deliberately adopts this full-information regime to isolate the surrogate-design question from the orthogonal question of partial cost observation. Extending the augmented surrogate to the *limited-information* setting — where only the executed pair’s cost is observed at training time, so that counterfactual expert–advice costs must be estimated (e.g., via off-policy evaluation, importance weighting, or bandit feedback) — is a natural next step and the primary direction of our future work.

8 Impact Statement

Learning-to-Defer with advice formalizes how a learned system can decide both which expert to consult and what additional information to provide. Its potential benefits include reduced reliance on the most expensive experts, cost-aware acquisition of sensitive or costly information, and more transparent allocation of decisions between automated and human reviewers. As with any deferral system, these benefits come with risks: the learned policy inherits the biases of its experts and of the cost structure chosen by the operator, and the acquisition of advice may involve sensitive information (e.g., escalating identity or payment data) whose use must respect applicable privacy, fairness, and compliance constraints. We view the present contribution as foundational — a calibrated surrogate with provable consistency guarantees — and encourage application-specific audits of experts, costs, and advice sources before deployment.

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719 A Algorithms, Visualization, Example

720 A.1 Notation Summary

721 Table 2 collects the main symbols used throughout the paper. We keep only the objects that are
722 central to the formulation and theory.

723 A.2 Visualization

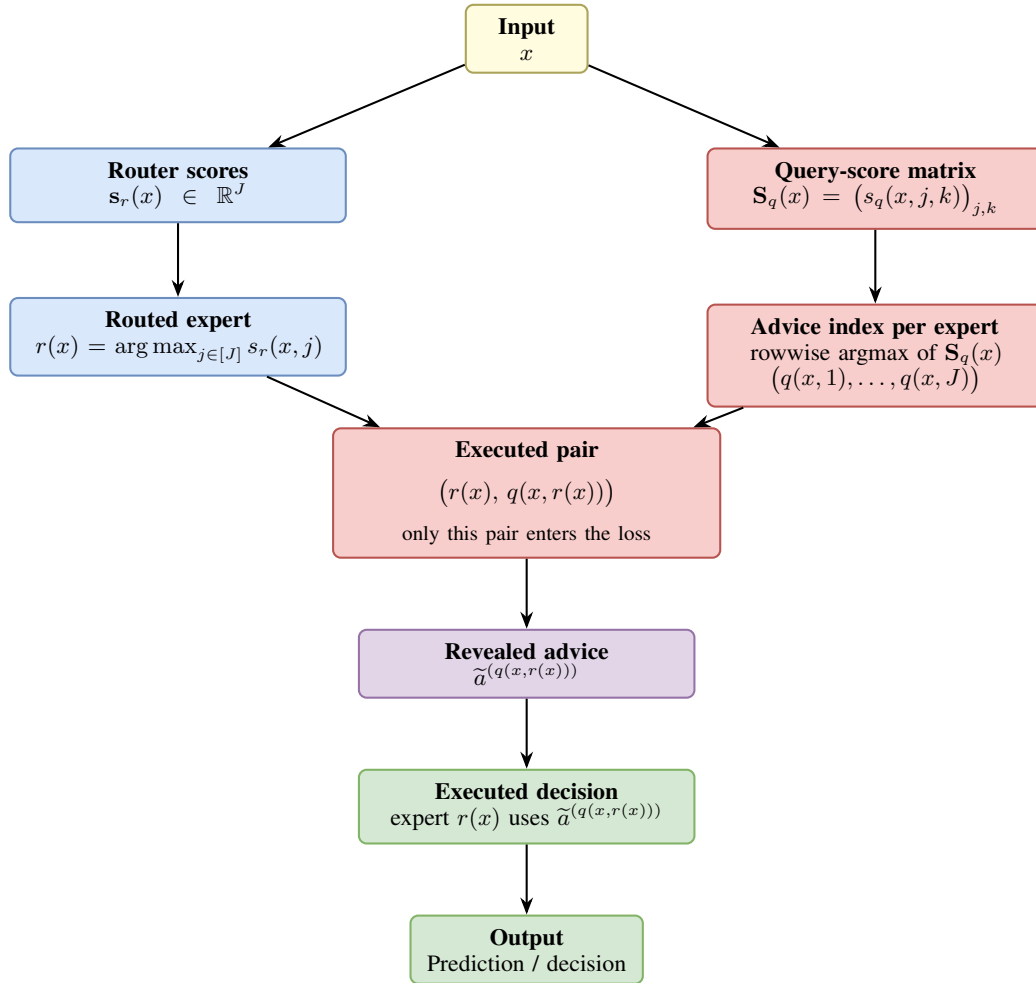


Figure 2: The deferral-advice protocol. From the input x , the router produces expert scores and the query model produces a $J \times (K + 1)$ query-score matrix. Taking the argmax along each row yields one advice index per expert, but only the routed row is ever executed: the protocol uses the pair $(r(x), q(x, r(x)))$, reveals the corresponding masked advice, and executes expert $r(x)$ under that advice. When $q(x, r(x)) = 0$, the revealed advice is the null masked advice $\tilde{a}^{(0)}$.

Table 2: Notation summary.

Symbol	Meaning
<i>Data, experts, and advice</i>	
$(X, Y) \sim \mathcal{D}$	Input-output random pair.
(x, y)	One realization of (X, Y) .
$A = (A^1, \dots, A^K)$	Advice random vector.
(x, a, y)	One realization of the full deferral-advice instance.
$[J] = \{1, \dots, J\}$	Expert index set.
$[K]_0 = \{0, 1, \dots, K\}$	Advice-action set, where $k = 0$ denotes no advice.
m_k	Masking operator that reveals at most advice source k .
$\tilde{a}^{(k)}$	Masked realized advice under action k .
$\tilde{A}^{(k)}$	Masked advice random variable under action k .
e_j	Expert predictor for expert j .
$\mathbf{e} = (e_1, \dots, e_J)$	Tuple of expert predictors.
ψ	Task loss.
β_j	Expert consultation cost for expert j .
γ_k	Advice acquisition cost for advice action k .
$c_j(x, y)$	Standard L2D cost of routing (x, y) to expert j .
$c_{j,k}(x, a, y)$	Deferral-advice cost of executing expert-advice pair (j, k) .
<i>Policies, actions, and risks</i>	
$r : \mathcal{X} \rightarrow [J]$	Router in standard Learning-to-Defer.
$q : \mathcal{X} \times [J] \rightarrow [K]_0$	Expert-conditional query function.
ℓ_{def}	Standard deferral loss.
$\ell_{\text{def-adv}}$	True deferral-advice loss.
$\mathcal{E}_{\ell_{\text{def}}}$	Population risk associated with ℓ_{def} .
$\mathcal{E}_{\ell_{\text{def-adv}}}$	Population risk associated with $\ell_{\text{def-adv}}$.
r^*	Bayes-optimal router.
q^*	Bayes-optimal query function.
$\Pi = [J] \times [K]_0$	Composite action space of executed expert-advice pairs.
$i \in \Pi$	Composite action index, typically identified with a pair (j, k) .
$\pi : \mathcal{X} \rightarrow \Pi$	Composite predictor.
<i>Surrogates and consistency quantities</i>	
$\mathcal{H}_r$	Hypothesis class for router scores.
$\mathcal{H}_q$	Hypothesis class for query scores.
$\mathcal{H}_\pi$	Hypothesis class for composite scores.
s_r	Router score function.
$\mathbf{s}_r(x)$	Router score vector over experts.
s_q	Query score function over expert-advice pairs (j, k) .
$\mathbf{S}_q(x)$	Query-score matrix with entries $s_q(x, j, k)$.
s_π	Composite score function.
$\mathbf{s}_\pi(x)$	Composite score vector over Π .
$\Phi_{\text{def-adv}}^{r/q}$	Separated router/query surrogate family.
$F(u, v)$	Profiled row summary in the separated-surrogate analysis.
$D(x, a, y)$	Action-independent offset in the composite-loss decomposition.
$w_i(x, a, y)$	Mismatch weight associated with composite action i .
Φ_{01}^τ	Comp-sum multiclass loss indexed by $\tau \geq 0$.
$\Phi_{\text{def-adv}}^{\text{aug}, \tau}$	Augmented deferral-advice surrogate.
Γ_τ	Transfer function for the base comp-sum loss on Π .
$\tilde{\Gamma}_\tau$	Transfer function in the augmented-surrogate consistency bound.

724 A.3 Worked Example of the True Loss

725 The following example traces the full evaluation of the deferral-advice loss on a single realization,
726 starting from the scoring functions and ending with the scalar loss value.

727 **Example 1** (Evaluating the true loss on one instance). *Consider $J = 3$ experts and $K = 2$ advice*
728 *sources, so the advice index set is $[K]_0 = \{0, 1, 2\}$.*

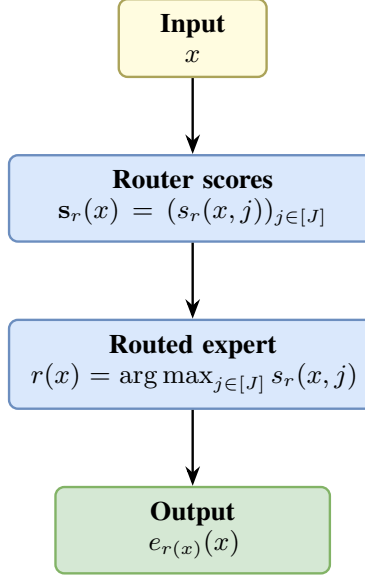


Figure 3: The standard Learning-to-Defer protocol. From the input x , the router produces one score per expert, routes to the highest-scoring expert $r(x)$, and outputs that expert’s prediction. Unlike the deferral-advice protocol in Figure 2, there is no query head, no advice action, and no executed expert–advice pair.

729 **Scores.** Suppose the router and query scoring functions produce, for a given input x ,

$$\mathbf{s}_r(x) = (0.8, 1.2, 0.5), \quad \mathbf{S}_q(x) = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.9 & 0.4 \\ 0.5 & 0.1 & 0.7 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

730 Each row of $\mathbf{S}_q(x)$ corresponds to one expert; each column to one advice action ($k = 0, 1, 2$).

731 **Policy decisions.** The router selects the expert with the highest routing score: $r(x) =$
 732 $\arg \max_{j \in [3]} s_r(x, j) = 2$. The query function computes one advice index per expert by taking
 733 the row-wise argmax of $\mathbf{S}_q(x)$:

$$q(x, 1) = 0, \quad q(x, 2) = 1, \quad q(x, 3) = 2.$$

734 Only the executed query matters: $k^* = q(x, r(x)) = q(x, 2) = 1$.

735 **Realized costs.** For this realization (x, a, y) , suppose the full cost table is

$$(c_{j,k}(x, a, y))_{j \in [3], k \in \{0,1,2\}} = \begin{pmatrix} 0.35 & 0.42 & 0.38 \\ 0.40 & 0.20 & 0.45 \\ 0.50 & 0.48 & 0.25 \end{pmatrix}.$$

736 Each entry includes task loss, expert fee, and advice fee.

737 **Loss evaluation.** The double indicator in Definition 2 selects the executed pair $(j, k) = (2, 1)$:

$$\ell_{\text{def-adv}}(r, q; x, a, y, \mathbf{e}) = c_{2,1}(x, a, y) = 0.20.$$

738 The counterfactual query decisions $q(x, 1) = 0$ and $q(x, 3) = 2$ play no role: the protocol only pays
 739 for the executed pair. The chosen pair is also the realized minimum of the cost table: comparing
 740 the row-wise minima $(0.35, 0.20, 0.25)$, expert 2 with advice $k = 1$ achieves the smallest cost. This
 741 is exactly the comparison performed by the Bayes policy in Lemma 3, applied at the level of one
 742 realization.

743 A.4 Algorithms

744 Algorithms 1, 3, and 2 summarize the two decision rules discussed in the paper. The first two are the
 745 proposed augmented approach: training from fully observed executed-pair costs, and deployment-
 746 time execution of the learned composite policy. Algorithm 2 then records the inference rule of
 747 the separated router/query parameterization, which is the natural sequential baseline analyzed in
 748 Section 4.4.

Algorithm 1 Training with the augmented surrogate

Require: Training set $\{(x_n, a_n, y_n)\}_{n=1}^N$, composite action set $\Pi = [J] \times [K]_0$, score model $s_\pi(\cdot, \cdot; \theta)$, comp-sum parameter $\tau \geq 0$, learning rate η

1: Initialize parameters θ

2: **repeat**

3: Sample a minibatch $B \subseteq \{1, \dots, N\}$

4: **for all** $n \in B$ **do**

5: **for all** $i = (j, k) \in \Pi$ **do**

6: Compute the realized cost $c_i(x_n, a_n, y_n) \equiv c_{j,k}(x_n, a_n, y_n)$

7: **end for**

8: Set $w_i(x_n, a_n, y_n) \leftarrow \max_{i' \in \Pi} c_{i'}(x_n, a_n, y_n) - c_i(x_n, a_n, y_n)$ for all $i \in \Pi$

9: Form the sample loss

$$L_n(\theta) \leftarrow \sum_{i \in \Pi} w_i(x_n, a_n, y_n) \Phi_{01}^\tau(\mathbf{s}_\pi(x_n; \theta), i)$$

10: **end for**

11: Update $\theta \leftarrow \theta - \eta \nabla_\theta (|B|^{-1} \sum_{n \in B} L_n(\theta))$

12: **until** convergence

13: **return** learned parameters $\hat{\theta}$

Algorithm 2 Inference with a separated router/query policy

Require: Input x , router score model $s_r(\cdot, \cdot)$, query score model $s_q(\cdot, \cdot, \cdot)$

1: Form the query-score matrix

$$\mathbf{S}_q(x) \leftarrow (s_q(x, j, k))_{j \in [J], k \in [K]_0} \in \mathbb{R}^{J \times (K+1)}$$

2: For each expert j , take the argmax along row j of $\mathbf{S}_q(x)$:

$$\hat{k}_j(x) \in \arg \max_{k \in [K]_0} s_q(x, j, k)$$

3: Compute the routed expert

$$\hat{j}(x) \in \arg \max_{j \in [J]} s_r(x, j)$$

4: Select the advice actually executed by the routed expert

$$\hat{k}(x) \leftarrow \hat{k}_{\hat{j}(x)}(x)$$

5: **if** $\hat{k}(x) = 0$ **then**

6: Set $\tilde{a}^{(\hat{k}(x))} \leftarrow \tilde{a}^{(0)}$

7: **else**

8: Acquire advice source $\hat{k}(x)$ and reveal $\tilde{a}^{(\hat{k}(x))}$

9: **end if**

10: Route x to expert $\hat{j}(x)$ and output $e_{\hat{j}(x)}(x, \tilde{a}^{(\hat{k}(x))})$

11: **return** routed expert $\hat{j}(x)$, executed advice $\hat{k}(x)$, and prediction $e_{\hat{j}(x)}(x, \tilde{a}^{(\hat{k}(x))})$

Algorithm 3 Inference with the learned composite policy

Require: Input x , learned score model $s_\pi(\cdot, \cdot; \hat{\theta})$

1: Select the composite action

$$(\hat{j}, \hat{k}) \leftarrow \arg \max_{(j,k) \in \Pi} s_\pi(x, (j, k); \hat{\theta})$$

2: **if** $\hat{k} = 0$ **then**

3: Set $\tilde{a}^{(\hat{k})} \leftarrow \tilde{a}^{(0)}$

4: **else**

5: Acquire advice source $\hat{k}$ and reveal the masked advice $\tilde{a}^{(\hat{k})}$

6: **end if**

7: Route x to expert $\hat{j}$ and output $e_{\hat{j}}(x, \tilde{a}^{(\hat{k})})$

8: **return** executed pair $\hat{\pi}(x) = (\hat{j}, \hat{k})$ and prediction $e_{\hat{j}}(x, \tilde{a}^{(\hat{k})})$

B Proofs

Example 2 (Running conditional-cost table). *Before turning to the formal derivations, it is useful to keep one concrete conditional cost table in mind. Fix an input x and consider*

$$\bar{C}(x) := (\mathbb{E}[c_{j,k}(X, A, Y) \mid X = x])_{j \in [3], k \in \{0,1,2\}} = \begin{pmatrix} 0.32 & 0.41 & 0.36 \\ 0.35 & 0.27 & 0.40 \\ 0.38 & 0.33 & 0.24 \end{pmatrix}.$$

Each row corresponds to an expert, each column to an advice action, and $k = 0$ denotes the no-advice option. The entries already include the entire executed cost: prediction error, expert fee, and advice fee. The query decision therefore acts row by row, because once the expert is fixed the learner still has to decide which advice source, if any, should be revealed to that expert. In this table, expert 1 is best with no advice, expert 2 is best with advice 1, and expert 3 is best with advice 2. The expert-wise Bayes query therefore keeps only the row minima

$$(0.32, 0.27, 0.24),$$

and the Bayes router then selects expert 3, because 0.24 is the smallest of these three values.

Standard Learning-to-Defer would reason differently. It would only compare the unaided column

$$(0.32, 0.35, 0.38),$$

and would therefore select expert 1. The richer protocol changes the decision because it allows each expert to be evaluated under its own best advice.

This example will be useful in the next three proofs. It illustrates the three main ideas: the best advice is expert-dependent, the router should compare experts only after this expert-wise query choice has been made, and the additional advice menu can strictly improve the achievable Bayes risk.

B.1 Proof of Lemma 3

Lemma 3 (Bayes-optimal deferral-advice policy). *There exists a measurable Bayes-optimal policy (r^*, q^*) that minimizes $\mathbb{E}[\ell_{\text{def-adv}}(r, q; X, A, Y, \mathbf{e})]$ over all measurable policies and, for $\mathbb{P}_X$ -a.e. x , can be chosen by smallest-index tie-breaking as follows:*

1. The Bayes query selects the best advice for each expert:

$$q^*(x, j) = \min_{k \in [K]_0} \arg \min \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x]. \quad (4)$$

2. The Bayes router selects the best optimally-advised expert:

$$r^*(x) = \min_{j \in [J]} \arg \min \mathbb{E}[c_{j,q^*(x,j)}(X, A, Y) \mid X = x]. \quad (5)$$

Proof The key point is that the protocol pays only for the executed pair $(r(x), q(x, r(x)))$. Once this is made explicit, the Bayes problem reduces to a pointwise comparison of the conditional expected costs of the finitely many expert-advice pairs.

Indeed, by Definition 2, the population risk of any policy (r, q) can be written as

$$\mathbb{E}[\ell_{\text{def-adv}}(r, q; X, A, Y, \mathbf{e})] = \mathbb{E} \left[\sum_{j \in [J]} \sum_{k \in [K]_0} c_{j,k}(X, A, Y) \mathbf{1}\{r(X) = j\} \mathbf{1}\{q(X, j) = k\} \right].$$

Since exactly one pair (j, k) is selected for each X , this simplifies to

$$\mathbb{E}[c_{r(X), q(X, r(X))}(X, A, Y)].$$

Thus the loss depends on the policy only through the executed pair. Applying the tower property of conditional expectation gives

$$\mathbb{E}[c_{r(X), q(X, r(X))}(X, A, Y)] = \mathbb{E}[\mathbb{E}[c_{r(X), q(X, r(X))}(X, A, Y) \mid X]].$$

Since $r(X)$ and $q(X, r(X))$ are deterministic functions of X , conditioning on X fixes the pair $(j, k) = (r(X), q(X, r(X)))$, and the inner expectation becomes $\mathbb{E}[c_{j,k}(X, A, Y) \mid X]$. It is therefore enough to minimize this conditional expectation pointwise in x .

781 For $\mathbb{P}_X$ -a.e. x , the optimization problem is

$$\min_{(j,k) \in [J] \times [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x].$$

782 Since $[J] \times [K]_0$ is a finite set and each $c_{j,k}$ is bounded (hence integrable), the conditional expectations
 783 $\mathbb{E}[c_{j,k}(X, A, Y) \mid X = x]$ are well-defined measurable functions of x . The minimum of finitely
 784 many measurable functions is measurable, and choosing the smallest index among the finitely many
 785 minimizers produces a measurable selector. This minimum can be decomposed as

$$\min_{(j,k) \in [J] \times [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x] = \min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x],$$

786 so the optimal decision decomposes in the same way. For each expert j , the smallest-index Bayes
 787 query is

$$q^*(x, j) = \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x].$$

788 Once these expert-wise best query values are fixed, the router compares the resulting optimally-
 789 advised costs and therefore satisfies

$$r^*(x) = \min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x] = \min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x].$$

790 This is exactly the pair of Bayes decisions stated in (4) and (5). Example 3 shows this decomposition
 791 concretely: the query first selects one minimizing column per row, and the router then compares only
 792 these row minima. Consequently, the Bayes risk is

$$\mathcal{E}_{\ell_{\text{def-adv}}}^* = \mathbb{E}\left[\min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X]\right].$$

793 ■

794 **Example 3** (Bayes query and Bayes router on a conditional cost table). *Fix an input x and suppose*
 795 *the conditional executed-pair costs are*

$$\bar{\mathbf{C}}(x) := (\mathbb{E}[c_{j,k}(X, A, Y) \mid X = x])_{j \in [3], k \in \{0,1,2\}} = \begin{pmatrix} 0.32 & 0.41 & 0.36 \\ 0.35 & 0.27 & 0.40 \\ 0.38 & 0.33 & 0.24 \end{pmatrix}.$$

796 *Each row corresponds to one expert, and each column to one advice action, with $k = 0$ denoting no*
 797 *advice. The Bayes query acts row by row: for expert 1, the smallest entry is 0.32, so $q^*(x, 1) = 0$;*
 798 *for expert 2, the smallest entry is 0.27, so $q^*(x, 2) = 1$; and for expert 3, the smallest entry is 0.24,*
 799 *so $q^*(x, 3) = 2$. Once these query decisions are fixed, the router does not compare the full rows*
 800 *anymore. It compares the best achievable cost of each expert,*

$$(\mathbb{E}[c_{1,q^*(x,1)} \mid X = x], \mathbb{E}[c_{2,q^*(x,2)} \mid X = x], \mathbb{E}[c_{3,q^*(x,3)} \mid X = x]) = (0.32, 0.27, 0.24),$$

801 and therefore selects

$$r^*(x) = 3.$$

802 *First determine, for each expert, which advice makes that expert cheapest; then route to the expert*
 803 *whose best-advised conditional cost is smallest.*

804 B.2 Proof of Lemma 4

805 **Lemma 4** (Advice acquisition condition). *Under the cost decomposition (3) and the smallest-index*
 806 *tie-breaking of Lemma 3, the Bayes-optimal query satisfies $q^*(x, j) \neq 0$ if and only if there exists*
 807 *$k \in [K]$ such that*

$$\mathbb{E}[\psi(e_j(X, \tilde{A}^{(0)}), Y) \mid X = x] - \mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X = x] > \gamma_k. \quad (6)$$

808 *When no source satisfies this condition, $q^*(x, j) = 0$ and the expert acts on x alone.*

809 **Proof** Once the expert is fixed, the only remaining decision is whether the gain from revealing
 810 advice justifies its acquisition cost. The smallest-index tie-breaking in Lemma 3 is what converts the
 811 resulting strict comparison with the no-advice option into an “if and only if” statement.

812 Fix an expert $j \in [J]$ and an input x . For every $k \in [K]_0$,

$$\mathbb{E}[c_{j,k}(X, A, Y) \mid X = x] = \mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X = x] + \beta_j + \gamma_k.$$

813 The expert fee β_j does not depend on the advice index and therefore plays no role in the comparison
814 across k . For a fixed expert, the smallest-index Bayes query of Lemma 3 thus solves

$$q^*(x, j) \in \arg \min_{k \in [K]_0} \left\{ \mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X = x] + \gamma_k \right\}.$$

815 The no-advice action corresponds to $k = 0$ and has cost

$$\mathbb{E}[\psi(e_j(X, \tilde{A}^{(0)}), Y) \mid X = x],$$

816 because $\gamma_0 = 0$. The policy therefore strictly prefers a queried advice source $k \geq 1$ to no advice
817 exactly when

$$\mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X = x] + \gamma_k < \mathbb{E}[\psi(e_j(X, \tilde{A}^{(0)}), Y) \mid X = x],$$

818 which rearranges to

$$\mathbb{E}[\psi(e_j(X, \tilde{A}^{(0)}), Y) \mid X = x] - \mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X = x] > \gamma_k.$$

819 This is exactly condition (6). Hence, if some $k \in [K]$ satisfies (6), then that queried action has strictly
820 smaller conditional total cost than $k = 0$, so the Bayes-optimal query cannot be 0. Conversely, if no
821 queried action satisfies (6), then every $k \in [K]$ has conditional total cost at least as large as that of
822 $k = 0$. Because ties are broken in favor of the smallest advice index and 0 is the smallest index in
823 $[K]_0$, the selected Bayes query is then $q^*(x, j) = 0$. This proves the lemma. ■

824 B.3 Proof of Lemma 5

825 **Lemma 5** (Deferral with advice dominates standard deferral). *If $\mathcal{E}_{\ell_{\text{def}}}^*$ denotes the Bayes risk of the*
826 *standard deferral setting of Section 3, then*

$$\mathcal{E}_{\ell_{\text{def-adv}}}^* \leq \mathcal{E}_{\ell_{\text{def}}}^*. \quad (7)$$

827 **Proof** The intuition is simple. Deferral with advice enlarges the menu of admissible actions, but it
828 never removes the standard L2D action. For any fixed expert, the best cost achievable with advice
829 can therefore only improve upon, or match, the unaided one. The content of the lemma is that this
830 row-wise comparison is exactly the one the Bayes router uses, so it lifts directly to the population
831 Bayes risk.

832 Fix an expert $j \in [J]$ and an input x in a full-measure set on which the relevant conditional
833 expectations are defined. Since the null action $k = 0$ belongs to $[K]_0$, minimizing over the larger
834 advice set immediately gives

$$\min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x] \leq \mathbb{E}[c_{j,0}(X, A, Y) \mid X = x]. \quad (14)$$

835 Expert j can always recover its unaided behavior by choosing $k = 0$.

836 Because (14) holds for every expert, it remains true after taking the outer minimum over $j \in [J]$:

$$\min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x] \leq \min_{j \in [J]} \mathbb{E}[c_{j,0}(X, A, Y) \mid X = x].$$

837 The left-hand side is precisely the conditional quantity minimized by the Bayes router of Lemma 3.
838 On the right-hand side, the action $k = 0$ coincides with standard Learning-to-Defer, so $c_{j,0}(x, a, y) =$
839 $c_j(x, y)$ and therefore

$$\mathbb{E}[c_{j,0}(X, A, Y) \mid X = x] = \mathbb{E}[c_j(X, Y) \mid X = x].$$

840 Integrating over X therefore yields

$$\mathcal{E}_{\ell_{\text{def-adv}}}^* = \mathbb{E} \left[\min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X] \right] \leq \mathbb{E} \left[\min_{j \in [J]} \mathbb{E}[c_j(X, Y) \mid X] \right] = \mathcal{E}_{\ell_{\text{def}}}^*,$$

841 which proves (7). ■

Remark 10 (Equality condition in Lemma 5). *The equality case must be stated at the level of the best available expert, not expert by expert. Define*

$$g_{\text{adv}}(x) := \min_{j \in [J]} \min_{k \in [K]_0} \mathbb{E}[c_{j,k}(X, A, Y) \mid X = x], \quad g_{\text{def}}(x) := \min_{j \in [J]} \mathbb{E}[c_{j,0}(X, A, Y) \mid X = x].$$

The first part of the proof shows that $g_{\text{adv}}(x) \leq g_{\text{def}}(x)$ for $\mathbb{P}_X$ -a.e. x . Equivalently, the gap

$$h(x) := g_{\text{def}}(x) - g_{\text{adv}}(x)$$

is nonnegative for $\mathbb{P}_X$ -a.e. x . Equality in (7),

$$\mathbb{E}[g_{\text{adv}}(X)] = \mathbb{E}[g_{\text{def}}(X)],$$

is therefore equivalent to $\mathbb{E}[h(X)] = 0$. Since $h(X) \geq 0$ almost surely, this happens if and only if

$$g_{\text{adv}}(x) = g_{\text{def}}(x) \quad \text{for } \mathbb{P}_X\text{-a.e. } x.$$

Advice may improve some non-selected experts without changing the Bayes risk. What matters is whether it lowers the smallest conditional cost available at x on a set of positive $\mathbb{P}_X$ -measure.

Example 4 (Lemma 5 on the running table). *Return to Example 2. The conditional cost table is*

$$\bar{\mathbf{C}}(x) = \begin{pmatrix} 0.32 & 0.41 & 0.36 \\ 0.35 & 0.27 & 0.40 \\ 0.38 & 0.33 & 0.24 \end{pmatrix}.$$

The Bayes query function is obtained row by row:

$$q^*(x, 1) = 0, \quad q^*(x, 2) = 1, \quad q^*(x, 3) = 2.$$

This means that expert 1 should act without advice, expert 2 should receive advice source 1, and expert 3 should receive advice source 2. The query decision therefore acts within each row: once the expert is fixed, it keeps only the advice action that makes that expert cheapest. After this expert-wise selection, the three rows reduce to the best-achievable cost vector

$$(\mathbb{E}[c_{1,q^*(x,1)} \mid X = x], \mathbb{E}[c_{2,q^*(x,2)} \mid X = x], \mathbb{E}[c_{3,q^*(x,3)} \mid X = x]) = (0.32, 0.27, 0.24).$$

The Bayes router therefore chooses

$$r^*(x) = 3,$$

because expert 3 now has the smallest achievable conditional cost.

For comparison, standard Learning-to-Defer only sees the no-advice column,

$$(\mathbb{E}[c_{1,0} \mid X = x], \mathbb{E}[c_{2,0} \mid X = x], \mathbb{E}[c_{3,0} \mid X = x]) = (0.32, 0.35, 0.38),$$

so its Bayes router would select expert 1. Denoting this standard router by r_{def}^ , we therefore have*

$$r_{\text{def}}^*(x) = 1 \quad \text{whereas} \quad r^*(x) = 3.$$

The reason is exactly the one formalized in Lemma 5: once the advice menu is available, each row can only stay the same or improve, because $k = 0$ remains feasible. In this example, the no-advice comparison vector

$$(0.32, 0.35, 0.38)$$

is replaced by the best-advised vector

$$(0.32, 0.27, 0.24).$$

This new vector is coordinatewise no larger, and it is strictly smaller in the second and third coordinates. What matters for the lemma is the smallest entry of these two vectors, because that entry determines the Bayes decision at x . Here the best achievable conditional cost drops from

$$g_{\text{def}}(x) = 0.32 \quad \text{to} \quad g_{\text{adv}}(x) = 0.24.$$

This is why the richer protocol changes the selected expert and strictly lowers the Bayes-risk contribution at this input. Had advice improved only a non-selected row while leaving the minimum at 0.32, the overall Bayes risk would have remained unchanged. This is precisely the distinction captured by the equality condition in Lemma 5.

870 B.4 Proof of Theorem 6

871 We begin from the exact binary deferral-advice loss and show how the usual logistic replacement
 872 recipe leads naturally to a separated router/query surrogate. We then identify the quantity that this
 873 surrogate actually compares after the query heads are optimized out: each expert row is compressed
 874 into a profiled summary $F(c_{j,0}, c_{j,1})$, not into its row minimum. The final step is to construct a
 875 bounded cost table on which these two comparisons disagree. This shows that the failure is not tied
 876 to one peculiar parameterization, but to the separated architecture itself.

877 We work in the binary setting $J = 2, K = 1$ introduced in Section 4.4. The surrogate (8) is
 878 parameterized by a cost transform ν , query functions G and U , the logistic pair Φ_0, Φ_1 , and router
 879 weights Ψ_1, Ψ_2 .

880 Let $s_r^b : \mathcal{X} \rightarrow \mathbb{R}$ be a binary router score and, for each expert $j \in \{1, 2\}$, let $s_{q_j^b} : \mathcal{X} \rightarrow \mathbb{R}$ be a binary
 881 query score. Write $\mathbf{s}_{q^b} = (s_{q_1^b}, s_{q_2^b})$. As in the main text, the decoded decisions are

$$r^b(x) = 1 + \mathbf{1}\{s_r^b(x) \geq 0\}, \quad q_j^b(x) = \mathbf{1}\{s_{q_j^b}(x) \geq 0\},$$

882 with $q^b(x, j) = q_j^b(x)$.

883 B.4.1 Proof of Proposition 11

884 **Proposition 11** (Natural separated construction). *For a realization (x, a, y) , the binary executed-pair*
 885 *loss admits the exact expansion*

$$\ell_{\text{def-adv}}^b(r^b, q^b; x, a, y, \mathbf{e}) := \sum_{j=1}^2 \sum_{k=0}^1 c_{j,k}(x, a, y) \mathbf{1}\{r^b(x) = j\} \mathbf{1}\{q_j^b(x) = k\}. \quad (15)$$

886 *Assigning logistic surrogate potentials to the decoded binary outcomes in the four terms of (15)*
 887 *yields the basic separated surrogate $\Phi_{\text{def-adv}}^{b,r/q}(s_r^b, \mathbf{s}_{q^b})$.*

888 **Proof** This proposition records the most direct surrogate construction suggested by the protocol
 889 itself. In the binary setting there are only four executed outcomes: route to expert 1 or expert 2, and
 890 for the selected expert either do not query or query the single advice source. Writing these four cases
 891 out explicitly makes it clear how the usual logistic replacement enters.

892 Fix (x, a, y) . Since exactly one expert and one binary advice decision are executed, expanding the
 893 two sums in (15) gives

$$\begin{aligned} \ell_{\text{def-adv}}^b(r^b, q^b; x, a, y, \mathbf{e}) &= c_{1,0}(x, a, y) \mathbf{1}\{s_r^b(x) < 0\} \mathbf{1}\{s_{q_1^b}(x) < 0\} \\ &\quad + c_{1,1}(x, a, y) \mathbf{1}\{s_r^b(x) < 0\} \mathbf{1}\{s_{q_1^b}(x) \geq 0\} \\ &\quad + c_{2,0}(x, a, y) \mathbf{1}\{s_r^b(x) \geq 0\} \mathbf{1}\{s_{q_2^b}(x) < 0\} \\ &\quad + c_{2,1}(x, a, y) \mathbf{1}\{s_r^b(x) \geq 0\} \mathbf{1}\{s_{q_2^b}(x) \geq 0\}. \end{aligned} \quad (16)$$

894 This is the exact executed-pair loss written in terms of the decoded binary scores.

895 Assigning logistic potentials to the decoded binary outcomes yields

$$\Phi_{\text{def-adv}}^{b,r/q}(s_r^b, \mathbf{s}_{q^b}) := \sum_{j=1}^2 \Phi_{j-1}(s_r^b(x)) \left[c_{j,0}(x, a, y) \Phi_0(s_{q_j^b}(x)) + c_{j,1}(x, a, y) \Phi_1(s_{q_j^b}(x)) \right].$$

896 This is exactly the basic logistic construction described in Section 4.4. This proves the natural
 897 separated construction. ■

898 **Regularity conditions for the larger separated family.** Equation (8) extends the basic construction
 899 by allowing monotone cost transforms, query reparameterizations, and flexible router weights. We
 900 impose the following conditions. They are not designed to make the counterexample artificially
 901 strong. They simply capture the regularity one would naturally ask of a well-behaved separated
 902 surrogate.

903 (N1) *Cost transform:* $\nu : [0, C] \rightarrow [0, \infty)$ is continuous, differentiable on $(0, C)$, and strictly
 904 increasing there, so $\nu'(c) > 0$ for $c \in (0, C)$.

905 (Q1) *Amplitude and reparameterization:* $G : \mathbb{R} \rightarrow (0, \infty)$ is continuous and strictly positive;
 906 $U : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, strictly increasing, and surjective.
 907 (Q2) *Well-posedness:* for every $(u, v) \in (0, C]^2$, $t \mapsto \nu(u) G(t) \Phi_0(U(t)) + \nu(v) G(t) \Phi_1(U(t))$ is
 908 coercive and strictly convex with a unique minimizer $t^*(u, v)$, and the map $(u, v) \mapsto t^*(u, v)$
 909 is continuous on $(0, C]^2$.

910 (P1) *Router weights:* $\Psi_1, \Psi_2 \in C^2(\mathbb{R})$ with $\Psi_j > 0$, $\Psi'_1 < 0$, $\Psi'_2 > 0$, $\Psi''_j > 0$.
 911 (P2) *Routing ratio:* $\rho(r) := -\Psi'_1(r)/\Psi'_2(r)$ is a strictly decreasing bijection $\mathbb{R} \rightarrow (0, +\infty)$ with
 912 $\rho(0) = 1$.

913 These conditions are satisfied, for instance, by the standard logistic (cross-entropy) surrogate with
 914 softmax router weights on the strictly positive cost regime used in the theorem below.

915 **Containment of the traditional surrogate.** The familiar construction $\Phi_{\text{def-adv}}^{b,r/q}$ is recovered by
 916 taking $\nu(c) = c$, $G(q) \equiv 1$, and $U(q) = q$, together with the usual logistic router weights. The
 917 theorem proved below therefore applies to a broad family of positive-cost separated models that still
 918 preserves the same separated router/query architecture. The next lemmas isolate the precise object
 919 this architecture optimizes.

920 **The expert summary.** The next three lemmas formalize the quantity introduced in the main
 921 text. They show that after profiling over the query score of expert j , the entire row
 922 $(c_{j,0}(x, a, y), c_{j,1}(x, a, y))$ is compressed into the scalar summary $F(c_{j,0}(x, a, y), c_{j,1}(x, a, y))$. The
 923 key point is that this summary remains strictly sensitive to the non-minimal entry of the row. This is
 924 precisely where the surrogate departs from the Bayes comparison.

925 **Definition 12** (Expert summary). *For $(u, v) \in [0, C]^2$, define*

$$F(u, v) := \inf_{t \in \mathbb{R}} [\nu(u) G(t) \Phi_0(U(t)) + \nu(v) G(t) \Phi_1(U(t))]. \quad (17)$$

926 B.4.2 Proof of Lemma 13

927 **Lemma 13** (Well-defined expert summary). *Under (N1)–(Q2), for every $(u, v) \in (0, C]^2$, the*
 928 *infimum in (17) is attained at a unique point $t^*(u, v) \in \mathbb{R}$.*

929 **Proof** Fix $(u, v) \in (0, C]^2$ and define

$$f(t; u, v) := \nu(u) G(t) \Phi_0(U(t)) + \nu(v) G(t) \Phi_1(U(t)).$$

930 By (Q2), $f(\cdot; u, v)$ is coercive and strictly convex on $\mathbb{R}$. Coercivity implies that every sublevel set is
 931 compact, hence a global minimizer exists. Strict convexity then implies that this minimizer is unique.
 932 ■

933 B.4.3 Proof of Lemma 14

934 **Lemma 14** (Expert summary after profiling). *Under (P1) and (Q2), for every fixed $s_r^b : \mathcal{X} \rightarrow \mathbb{R}$ and*
 935 *every realization (x, a, y) whose four binary costs lie in $(0, C]$,*

$$\begin{aligned} \inf_{s_{q_1^b}, s_{q_2^b}} \Phi_{\text{def-adv}}^{\nu,r/q}(s_r^b, \mathbf{s}_{q^b}) &= \Psi_1(s_r^b(x)) F(c_{1,0}(x, a, y), c_{1,1}(x, a, y)) \\ &\quad + \Psi_2(s_r^b(x)) F(c_{2,0}(x, a, y), c_{2,1}(x, a, y)). \end{aligned} \quad (18)$$

936 *The profiled query minimizers are $t_j^* = t^*(c_{j,0}(x, a, y), c_{j,1}(x, a, y))$, hence they are independent of*
 937 *the router score s_r^b .*

938 **Proof** Fix (x, a, y) and write $t_r = s_r^b(x)$, $t_1 = s_{q_1^b}(x)$, and $t_2 = s_{q_2^b}(x)$. By (8),

$$\begin{aligned} \Phi_{\text{def-adv}}^{\nu,r/q}(s_r^b, \mathbf{s}_{q^b}) &= \Psi_1(t_r) \left[\nu(c_{1,0}(x, a, y)) G(t_1) \Phi_0(U(t_1)) \right. \\ &\quad \left. + \nu(c_{1,1}(x, a, y)) G(t_1) \Phi_1(U(t_1)) \right] \\ &\quad + \Psi_2(t_r) \left[\nu(c_{2,0}(x, a, y)) G(t_2) \Phi_0(U(t_2)) \right. \\ &\quad \left. + \nu(c_{2,1}(x, a, y)) G(t_2) \Phi_1(U(t_2)) \right]. \end{aligned}$$

939 The score t_1 appears only in the first bracket and t_2 appears only in the second. Because $\Psi_1(t_r)$ and
 940 $\Psi_2(t_r)$ are strictly positive by (P1), they factor through the infima over t_1 and t_2 . Therefore,

$$\begin{aligned} & \inf_{s_{q_1^b}, s_{q_2^b}} \Phi_{\text{def-adv}}^{\nu, r/q}(s_r^b, s_{q^b}) \\ &= \Psi_1(t_r) \inf_{t_1 \in \mathbb{R}} \left[\nu(c_{1,0}(x, a, y)) G(t_1) \Phi_0(U(t_1)) + \nu(c_{1,1}(x, a, y)) G(t_1) \Phi_1(U(t_1)) \right] \\ & \quad + \Psi_2(t_r) \inf_{t_2 \in \mathbb{R}} \left[\nu(c_{2,0}(x, a, y)) G(t_2) \Phi_0(U(t_2)) + \nu(c_{2,1}(x, a, y)) G(t_2) \Phi_1(U(t_2)) \right]. \end{aligned}$$

941 Each infimum is exactly an instance of the expert summary (17). This gives (18). The characterization
 942 of the minimizers follows directly from Lemma 13. ■

943 B.4.4 Proof of Lemma 15

944 **Lemma 15** (Continuity and strict monotonicity of the expert summary). *Under (N1)–(Q2), the map*
 945 *$F : (0, C]^2 \rightarrow (0, \infty)$ is continuous. Moreover, for every fixed $u \in (0, C)$, the map $v \mapsto F(u, v)$ is*
 946 *differentiable on $(0, C)$ and satisfies*

$$\frac{\partial F}{\partial v}(u, v) = \nu'(v) G(t^*(u, v)) \Phi_1(U(t^*(u, v))) > 0. \quad (19)$$

947 *In particular, for every fixed $u \in (0, C)$, the function $v \mapsto F(u, v)$ is strictly increasing on $(0, C)$.*

948 **Proof** Define

$$f(t; u, v) := \nu(u) G(t) \Phi_0(U(t)) + \nu(v) G(t) \Phi_1(U(t)).$$

949 By (N1) and (Q1), the map $(t, u, v) \mapsto f(t; u, v)$ is continuous. By Lemma 13, each pair $(u, v) \in$
 950 $(0, C]^2$ has a unique minimizer $t^*(u, v)$, and the continuity of $(u, v) \mapsto t^*(u, v)$ is part of (Q2).
 951 Hence

$$F(u, v) = f(t^*(u, v); u, v)$$

952 is continuous. Since $\nu(u) > 0$ and $\nu(v) > 0$ on $(0, C]$, and since G, Φ_0, Φ_1 are strictly positive, the
 953 attained value is also strictly positive.

954 Now fix $u \in (0, C)$ and $v \in (0, C)$. Since the minimizer is unique, Danskin's theorem applies to the
 955 function $v \mapsto \inf_{t \in \mathbb{R}} f(t; u, v) = F(u, v)$ and yields

$$\frac{\partial F}{\partial v}(u, v) = \frac{\partial}{\partial v} f(t; u, v) \Big|_{t=t^*(u, v)}.$$

956 Differentiating f with respect to v gives

$$\frac{\partial}{\partial v} f(t; u, v) = \nu'(v) G(t) \Phi_1(U(t)).$$

957 Substituting $t = t^*(u, v)$ proves (19). Every factor on the right-hand side is strictly positive:
 958 $\nu'(v) > 0$ by (N1), $G(t^*(u, v)) > 0$ by (Q1), and $\Phi_1(z) = \log(1 + e^z) > 0$ for every $z \in \mathbb{R}$.
 959 Therefore $\partial F / \partial v(u, v) > 0$, so $F(u, \cdot)$ is strictly increasing. ■

960 Lemmas 14 and 15 are the structural core of the impossibility result. Once the query scores are
 961 profiled out, expert j no longer appears through its row minimum. It appears through the scalar
 962 summary $F(c_{j,0}(x, a, y), c_{j,1}(x, a, y))$, and this summary depends strictly on the non-minimal entry.
 963 Bayes and the surrogate are therefore comparing different objects.

964 B.4.5 Proof of Lemma 16

965 **Lemma 16** (Router minimizer). *Under (P1)–(P2), for any $A, B > 0$ the function $t \mapsto A \Psi_1(t) +$*
 966 *$B \Psi_2(t)$ has a unique minimizer $t_r^*(A, B)$ satisfying*

$$A > B \implies t_r^* > 0, \quad A = B \implies t_r^* = 0, \quad A < B \implies t_r^* < 0.$$

967 *That is, the router selects the expert with the smaller profiled summary.*

968 **Proof** Strict convexity follows from (P1), since

$$\frac{d^2}{dt^2} [A \Psi_1(t) + B \Psi_2(t)] = A \Psi_1''(t) + B \Psi_2''(t) > 0.$$

969 Hence there is at most one minimizer. Differentiating the objective and setting the derivative to zero
970 yields

$$A \Psi_1'(t_r^*) + B \Psi_2'(t_r^*) = 0,$$

971 or equivalently

$$-\frac{\Psi_1'(t_r^*)}{\Psi_2'(t_r^*)} = \frac{B}{A}.$$

972 The left-hand side is $\rho(t_r^*)$, which is a strictly decreasing bijection from $\mathbb{R}$ to $(0, \infty)$ by (P2); since
973 $B/A \in (0, \infty)$, this FOC admits a unique solution, and by strict convexity any critical point of a C^2
974 strictly convex function is its global minimizer. Since $\rho(0) = 1$, we obtain

$$\frac{B}{A} < 1 \iff t_r^*(A, B) > 0, \quad \frac{B}{A} = 1 \iff t_r^*(A, B) = 0, \quad \frac{B}{A} > 1 \iff t_r^*(A, B) < 0.$$

975 This is exactly the stated sign characterization. ■

976 B.4.6 Completion of the proof of Theorem 6

977 **Theorem 6** (Broad Bayes inconsistency of separated router/query surrogates). *Under mild regularity*
978 *conditions on $\nu, G, U, \Psi_1, \Psi_2$ (Appendix B.4), for every $b \in (0, C)$ and $\varepsilon \in (0, C - b)$, there exists*
979 *$\delta \in (0, b)$ such that the pointwise conditional cost table*

$$\bar{\mathbf{C}}(x) = \begin{pmatrix} b - \delta & C \\ b & b + \varepsilon \end{pmatrix} \quad (10)$$

980 *has Bayes-optimal expert $j^* = 1$, while the unique minimizer of the pointwise surrogate (8) decodes*
981 *to expert 2. In particular, the entire family (8) is not Bayes consistent for the true loss.*

982 **Proof** Fix $b \in (0, C)$ and $\varepsilon \in (0, C - b)$. We will construct a bounded cost table on which Bayes
983 and the separated surrogate rank the two experts in opposite orders. The reason this can happen is
984 the one identified in the main text: Bayes compares row minima, whereas the separated surrogate
985 compares profiled row summaries.

986 Because $C > b + \varepsilon$ and $v \mapsto F(b, v)$ is strictly increasing by Lemma 15,

$$F(b, C) > F(b, b + \varepsilon). \quad (20)$$

987 The profile of the first row is therefore already larger than that of the second row when both rows
988 have the same minimum value b . Since $(u, v) \mapsto F(u, v)$ is continuous, we can perturb the first row
989 minimum slightly downward without destroying this strict inequality. More precisely, there exists
990 $\delta \in (0, b)$ such that

$$F(b - \delta, C) > F(b, b + \varepsilon). \quad (21)$$

991 Now consider the cost table (10). Its row minima are

$$\min\{b - \delta, C\} = b - \delta, \quad \min\{b, b + \varepsilon\} = b.$$

992 Since $\delta > 0$, Bayes compares these two values and selects expert 1.

993 The separated surrogate behaves differently. By Lemma 14, profiling out the two query scores reduces
994 the pointwise objective to

$$\tilde{\Phi}_{\text{def-adv}}^{\nu, r/q}(t) = F(b - \delta, C) \Psi_1(t) + F(b, b + \varepsilon) \Psi_2(t).$$

995 Define

$$A := F(b - \delta, C), \quad B := F(b, b + \varepsilon).$$

996 By (21), we have $A > B$. Lemma 16 then implies that the profiled router objective has a unique
997 minimizer $t_r^*(A, B) > 0$. Under the decoding rule

$$r^b(x) = 1 + \mathbf{1}\{s_r^b(x) \geq 0\},$$

998 this router minimizer decodes to expert 2.

999 To conclude, note that the query minimizers are also unique: for each row, Lemma 13 gives a unique
 1000 minimizer $t^*(c_{j,0}, c_{j,1})$, and Lemma 14 shows that these minimizers are independent of the router
 1001 score. Hence the full pointwise surrogate has a unique minimizer

$$(t_r^*(A, B), t^*(b - \delta, C), t^*(b, b + \varepsilon)),$$

1002 and this minimizer decodes to expert 2 even though Bayes selects expert 1. Thus the separated
 1003 surrogate fails the defining pointwise criterion for Bayes consistency. ■

1004 The counterexample already appears within a bounded cost table, so the failure does not rely on
 1005 any singular limiting construction. Nor does it disappear once the surrogate family is made more
 1006 flexible: the contradiction persists under monotone cost transforms, positive query amplitudes, smooth
 1007 reparameterizations, and well-behaved router weights. The obstruction is more basic. The separated
 1008 architecture compresses each row before routing, whereas the Bayes rule compares rows only through
 1009 their minima.

1010 B.5 Proof of the Sequential–Composite Equivalence

1011 **Proposition 17** (Sequential–composite equivalence). *For every sequential policy (r, q) , the composite*
 1012 *predictor $\pi(x) = (r(x), q(x, r(x)))$ incurs the same realized loss as (r, q) under $\ell_{\text{def-adv}}$. Conversely,*
 1013 *every composite predictor $\pi : \mathcal{X} \rightarrow \Pi$ can be implemented by a sequential policy with the same*
 1014 *realized loss. Consequently, the sequential and composite formulations induce the same population*
 1015 *risks and therefore share the same Bayes-optimal decisions.*

1016 **Proof** The proof is simply an identification of the object that the loss depends on. Although the
 1017 protocol is written sequentially, the realized loss only sees the executed pair consisting of the routed
 1018 expert and the advice ultimately shown to that expert.

1019 Let (r, q) be any sequential policy and define $\pi(x) = (r(x), q(x, r(x)))$. By Definition 2,

$$\ell_{\text{def-adv}}(r, q; x, a, y, \mathbf{e}) = c_{r(x), q(x, r(x))}(x, a, y) = c_{\pi_1(x), \pi_2(x)}(x, a, y) = \ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e}),$$

1020 where $\pi(x) = (\pi_1(x), \pi_2(x))$. Thus every sequential policy induces a composite predictor with
 1021 exactly the same realized loss at every input.

1022 Conversely, let $\pi : \mathcal{X} \rightarrow \Pi$ and write $\pi(x) = (\pi_1(x), \pi_2(x))$. To implement the same executed pair
 1023 sequentially, define

$$r(x) = \pi_1(x).$$

1024 The query function only matters at the routed expert, so it is enough to require

$$q(x, r(x)) = \pi_2(x).$$

1025 One convenient choice is to set $q(x, j) = \pi_2(x)$ for every $j \in [J]$. Then $q(x, r(x)) = \pi_2(x)$ and
 1026 therefore

$$\ell_{\text{def-adv}}(r, q; x, a, y, \mathbf{e}) = c_{\pi_1(x), \pi_2(x)}(x, a, y) = \ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e}).$$

1027 So every composite predictor can be realized by a sequential policy with the same pointwise loss.

1028 Since the two constructions agree pointwise in both directions, the sequential and composite formu-
 1029 lations induce exactly the same set of achievable realized losses. Their population risk functionals
 1030 therefore have the same infima and the same Bayes-optimal decisions. ■

1031 B.6 Proof of Proposition 18

1032 **Proposition 18** (Mismatch decomposition). *For every composite predictor $\pi : \mathcal{X} \rightarrow \Pi$ and every*
 1033 *realization (x, a, y) ,*

$$\ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e}) = D(x, a, y) + \sum_{i \in \Pi} w_i(x, a, y) \mathbf{1}\{\pi(x) \neq i\},$$

1034 where $M(x, a, y) = \max_{t \in \Pi} c_t(x, a, y)$, $w_i(x, a, y) = M(x, a, y) - c_i(x, a, y) \geq 0$, and
 1035 $D(x, a, y) = \sum_{i \in \Pi} c_i(x, a, y) - (|\Pi| - 1) M(x, a, y)$.

1036 **Proof** The decomposition is easiest to see by fixing the composite action that is actually executed.
 1037 Let

$$i_\pi := \pi(x) \in \Pi.$$

1038 Since the predictor selects exactly one composite action, the true loss is just the realized cost of that
 1039 action:

$$\ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e}) = c_{i_\pi}(x, a, y).$$

1040 Now expand the mismatch term. The indicator $\mathbf{1}\{\pi(x) \neq i\}$ vanishes only at the executed action
 1041 $i = i_\pi$, so the sum keeps every weight except the one attached to the chosen pair:

$$\begin{aligned} \sum_{i \in \Pi} w_i \mathbf{1}\{\pi(x) \neq i\} &= \sum_{i \neq i_\pi} w_i = \sum_{i \neq i_\pi} (M - c_i) \\ &= (|\Pi| - 1) M - \sum_{i \neq i_\pi} c_i. \end{aligned}$$

1042 Adding the offset

$$D = \sum_{i \in \Pi} c_i - (|\Pi| - 1) M$$

1043 therefore gives

$$D + \sum_{i \in \Pi} w_i \mathbf{1}\{\pi(x) \neq i\} = \sum_{i \in \Pi} c_i - (|\Pi| - 1) M + (|\Pi| - 1) M - \sum_{i \neq i_\pi} c_i = c_{i_\pi},$$

1044 which is exactly $\ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e})$. This proves the decomposition. ■

1045 B.7 Proof of Lemma 7

1046 **Proof** This lemma is a derivation rather than a separate statistical claim. Once the true loss is written
 1047 over the composite action space, the only action-dependent part is the weighted mismatch term. The
 1048 augmented surrogate is obtained by replacing each mismatch indicator with a multiclass surrogate
 1049 on Π .

1050 Indeed, Proposition 18 shows that

$$\ell_{\text{def-adv}}(\pi; x, a, y, \mathbf{e}) = D(x, a, y) + \sum_{i \in \Pi} w_i(x, a, y) \mathbf{1}\{\pi(x) \neq i\},$$

1051 where $D(x, a, y)$ does not depend on the chosen action. Thus, for surrogate design, the relevant
 1052 object is

$$\sum_{i \in \Pi} w_i(x, a, y) \mathbf{1}\{\pi(x) \neq i\}.$$

1053 The comp-sum family (Mao et al., 2023b) provides a multiclass surrogate for prediction on the
 1054 composite label space Π . Applying it label by label to the weighted mismatch term leads directly to

$$\Phi_{\text{def-adv}}^{\text{aug}, \tau}(s_\pi; x, a, y, \mathbf{e}) := \sum_{i \in \Pi} w_i(x, a, y) \Phi_{01}^\tau(s_\pi(x), i),$$

1055 which is exactly (12). In other words, the augmented surrogate is the comp-sum analogue of the
 1056 weighted mismatch representation of the true loss on the composite action space. ■

1057 B.8 Proof of Theorem 8

1058 The proof proceeds by reducing the augmented objective to an ordinary multiclass problem on the
 1059 composite action space Π , with example-dependent weights absorbed into a new data distribution.

1060 B.8.1 Proof of Lemma 19

1061 **Lemma 19** (Weighted multiclass reduction). *Let $\mathbf{w}(x) := (\mathbb{E}[w_i(X, A, Y) \mid X = x])_{i \in \Pi}$, and define*

$$d(x) := \mathbb{E}[D(X, A, Y) \mid X = x].$$

1062 *For every $i \in \Pi$, write*

$$\bar{w}_i(x) := \mathbb{E}[w_i(X, A, Y) \mid X = x].$$

1063 Define a probability vector $p(x) = (p_i(x))_{i \in \Pi}$ on Π by

$$p_i(x) := \begin{cases} \bar{w}_i(x)/\|\mathbf{w}(x)\|_1, & \|\mathbf{w}(x)\|_1 > 0, \\ 1/|\Pi|, & \|\mathbf{w}(x)\|_1 = 0, \end{cases} \quad i \in \Pi,$$

1064 and, for any composite predictor π and any score function s_π , define the conditional multiclass risks

$$C_{01}^p(\pi, x) := \sum_{i \in \Pi} p_i(x) \mathbf{1}\{\pi(x) \neq i\}, \quad C_{\Phi_{01}^\tau}^p(s_\pi, x) := \sum_{i \in \Pi} p_i(x) \Phi_{01}^\tau(s_\pi(x), i).$$

1065 Assume $\mathbb{E}[\|\mathbf{w}(X)\|_1] > 0$, and define a probability distribution $\tilde{\mathcal{D}}$ on $\mathcal{X} \times \Pi$ by

$$\tilde{\mathbb{P}}(X \in B, I = i) := \frac{1}{\mathbb{E}[\|\mathbf{w}(X)\|_1]} \mathbb{E}[\bar{w}_i(X) \mathbf{1}\{X \in B\}] \quad (22)$$

1066 for every measurable set $B \subseteq \mathcal{X}$ and every $i \in \Pi$. Let $\ell_{01}(\pi; x, i) := \mathbf{1}\{\pi(x) \neq i\}$ denote the
1067 multiclass 0–1 loss on Π . Then, for every scoring function $s_\pi \in \mathcal{H}_\pi$ and its decoded predictor
1068 $\pi(x) = \arg \max_{i \in \Pi} s_\pi(x, i)$,

$$C_{\ell_{\text{def-adv}}}(\pi, x) = d(x) + \|\mathbf{w}(x)\|_1 C_{01}^p(\pi, x), \quad (23)$$

$$C_{\Phi_{\text{aug}, \tau}}(s_\pi, x) = \|\mathbf{w}(x)\|_1 C_{\Phi_{01}^\tau}^p(s_\pi, x), \quad (24)$$

1069 where

$$C_{\ell_{\text{def-adv}}}(\pi, x) := \mathbb{E}[\ell_{\text{def-adv}}(\pi; X, A, Y, \mathbf{e}) \mid X = x]$$

1070 and

$$C_{\Phi_{\text{aug}, \tau}}(s_\pi, x) := \mathbb{E}[\Phi_{\text{def-adv}}^{\text{aug}, \tau}(s_\pi; X, A, Y, \mathbf{e}) \mid X = x].$$

1071 Consequently,

$$\mathcal{E}_{\Phi_{\text{aug}, \tau}}(s_\pi) = \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{E}_{\Phi_{01}^\tau}(s_\pi; \tilde{\mathcal{D}}), \quad (25)$$

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\pi) = \mathbb{E}[D(X, A, Y)] + \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{E}_{\ell_{01}}(\pi; \tilde{\mathcal{D}}). \quad (26)$$

1072 Moreover, the best-in-class risks and minimizability gaps satisfy

$$\mathcal{E}_{\Phi_{\text{aug}, \tau}}^*(\mathcal{H}_\pi) = \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{E}_{\Phi_{01}^\tau}^*(\mathcal{H}_\pi; \tilde{\mathcal{D}}), \quad (27)$$

$$\mathcal{U}_{\Phi_{\text{aug}, \tau}}(\mathcal{H}_\pi) = \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{U}_{\Phi_{01}^\tau}(\mathcal{H}_\pi; \tilde{\mathcal{D}}), \quad (28)$$

$$\mathcal{E}_{\ell_{\text{def-adv}}}^B(\mathcal{H}_\pi) = \mathbb{E}[D(X, A, Y)] + \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{E}_{\ell_{01}}^B(\mathcal{H}_\pi; \tilde{\mathcal{D}}), \quad (29)$$

$$\mathcal{U}_{\ell_{\text{def-adv}}}(\mathcal{H}_\pi) = \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{U}_{\ell_{01}}(\mathcal{H}_\pi; \tilde{\mathcal{D}}). \quad (30)$$

1073 **Proof** The point of the lemma is that, once we condition on $X = x$, the deferral-advice problem
1074 becomes an ordinary multiclass problem on the composite action space Π , up to an additive offset
1075 and a positive scale.

1076 Fix x . By Proposition 18,

$$\ell_{\text{def-adv}}(\pi; X, A, Y, \mathbf{e}) = D(X, A, Y) + \sum_{i \in \Pi} w_i(X, A, Y) \mathbf{1}\{\pi(X) \neq i\}.$$

1077 Taking the conditional expectation given $X = x$ gives

$$C_{\ell_{\text{def-adv}}}(\pi, x) = d(x) + \sum_{i \in \Pi} \bar{w}_i(x) \mathbf{1}\{\pi(x) \neq i\}.$$

1078 If $\|\mathbf{w}(x)\|_1 = 0$, then every $\bar{w}_i(x)$ vanishes, so the action-dependent term disappears and

$$C_{\ell_{\text{def-adv}}}(\pi, x) = d(x).$$

1079 Because the multiplier $\|\mathbf{w}(x)\|_1$ is also zero, both (23) and (24) hold automatically in this degenerate
1080 case, regardless of the arbitrary choice of $p(x)$.

1081 Assume now that $\|\mathbf{w}(x)\|_1 > 0$. Then $\bar{w}_i(x) = \|\mathbf{w}(x)\|_1 p_i(x)$ for every $i \in \Pi$, and therefore

$$C_{\ell_{\text{def-adv}}}(\pi, x) = d(x) + \|\mathbf{w}(x)\|_1 \sum_{i \in \Pi} p_i(x) \mathbf{1}\{\pi(x) \neq i\},$$

1082 which is exactly (23).

1083 The same normalization applies to the surrogate:

$$C_{\Phi^{\text{aug},\tau}(s_\pi, x) = \sum_{i \in \Pi} \bar{w}_i(x) \Phi_{01}^\tau(s_\pi(x), i) = \|\mathbf{w}(x)\|_1 \sum_{i \in \Pi} p_i(x) \Phi_{01}^\tau(s_\pi(x), i),$$

1084 which proves (24).

1085 Thus, for each fixed input x , the conditional deferral-advice problem is exactly a weighted multiclass
1086 problem on Π , up to the action-independent offset $d(x)$.

1087 To pass from this conditional reduction to a population statement, we average these weighted
1088 multiclass problems over X . Since $\bar{w}_i(X) \geq 0$ for all $i \in \Pi$, the right-hand side of (22) defines a
1089 finite non-negative measure on $\mathcal{X} \times \Pi$. Its total mass is

$$\sum_{i \in \Pi} \tilde{\mathbb{P}}(X \in \mathcal{X}, I = i) = \frac{1}{\mathbb{E}[\|\mathbf{w}(X)\|_1]} \sum_{i \in \Pi} \mathbb{E}[\bar{w}_i(X)] = \frac{1}{\mathbb{E}[\|\mathbf{w}(X)\|_1]} \mathbb{E}[\|\mathbf{w}(X)\|_1] = 1,$$

1090 so $\tilde{\mathcal{D}}$ is a probability distribution.

1091 Integrating (24) over X yields

$$\mathcal{E}_{\Phi^{\text{aug},\tau}(s_\pi) = \mathbb{E}\left[\sum_{i \in \Pi} \bar{w}_i(X) \Phi_{01}^\tau(s_\pi(X), i)\right] = \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{E}_{\Phi_{01}^\tau}(s_\pi; \tilde{\mathcal{D}}),$$

1092 which is (25). Likewise, (23) gives

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\pi) = \mathbb{E}[D(X, A, Y)] + \mathbb{E}[\|\mathbf{w}(X)\|_1] \mathcal{E}_{\ell_{01}}(\pi; \tilde{\mathcal{D}}),$$

1093 which is (26).

1094 The remaining identities follow from the same affine scaling. Taking infima over $s_\pi \in \mathcal{H}_\pi$ or over
1095 the induced predictors gives (27) and (29). Since the minimizability gaps are defined through these
1096 same best-in-class risks, the same argument also yields (28) and (30). ■

1097 **Theorem 8** ($\mathcal{H}_\pi$ -consistency of the augmented surrogate). *Assume that $\mathcal{H}_\pi$ satisfies the standard
1098 symmetry and completeness conditions required by the comp-sum multiclass $\mathcal{H}_\pi$ -consistency theorem
1099 for Φ_{01}^τ on the label set Π . For every $\tau \geq 0$, the augmented surrogate $\Phi_{\text{def-adv}}^{\text{aug},\tau}$ satisfies an $\mathcal{H}_\pi$ -
1100 consistency bound with respect to $\ell_{\text{def-adv}}$. Specifically, for every $s_\pi \in \mathcal{H}_\pi$ and its induced composite
1101 predictor π ,*

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\pi) - \mathcal{E}_{\ell_{\text{def-adv}}}^B(\mathcal{H}_\pi) + \mathcal{U}_{\ell_{\text{def-adv}}}(\mathcal{H}_\pi) \leq \tilde{\Gamma}_\tau \left(\mathcal{E}_{\Phi_{\text{def-adv}}^{\text{aug},\tau}(s_\pi) - \mathcal{E}_{\Phi_{\text{def-adv}}^*}^{\text{aug},\tau}(\mathcal{H}_\pi) + \mathcal{U}_{\Phi_{\text{def-adv}}^{\text{aug},\tau}}(\mathcal{H}_\pi) \right). \quad (13)$$

1102 With $\tilde{\Gamma}_\tau(u) := \mathbb{E}[\|\mathbf{w}(X)\|_1] \Gamma_\tau\left(\frac{u}{\mathbb{E}[\|\mathbf{w}(X)\|_1]}\right)$, where Γ_τ is the non-negative, non-decreasing,
1103 concave transfer function for Φ_{01}^τ on Π , and $\mathbf{w}(x) := (\mathbb{E}[w_i(X, A, Y) \mid X = x])_{i \in \Pi}$.

1104 **Proof** If $\mathbb{E}[\|\mathbf{w}(X)\|_1] = 0$, then $\bar{w}_i(X) = 0$ almost surely for every $i \in \Pi$. By Proposition 18,
1105 the true risk then reduces to the expectation of the action-independent term $D(X, A, Y)$, so every
1106 predictor is optimal and the theorem is trivial. We therefore assume $\mathbb{E}[\|\mathbf{w}(X)\|_1] > 0$.

1107 The key idea is to compare the surrogate and the true loss at the level where the Bayes policy itself
1108 is defined, namely conditionally on the input x . Lemma 19 shows that, once $X = x$ is fixed, both
1109 the true deferral-advice risk and the augmented surrogate reduce to weighted multiclass risks on the
1110 composite action space Π :

$$C_{\ell_{\text{def-adv}}}(\pi, x) = d(x) + \|\mathbf{w}(x)\|_1 C_{01}^p(\pi, x), \quad C_{\Phi^{\text{aug},\tau}}(s_\pi, x) = \|\mathbf{w}(x)\|_1 C_{\Phi_{01}^\tau}^p(s_\pi, x).$$

1111 The additive term $d(x)$ is independent of the chosen composite action, so it plays no role in the
1112 comparison between policies. What matters is that the action-dependent part of the true loss and the
1113 surrogate are governed by the same conditional weight vector $p(x)$.

1114 It is therefore natural to look first at the conditional excess risk. For each input x , define

$$\Delta C_{\ell_{\text{def-adv}}}(\pi, x) := C_{\ell_{\text{def-adv}}}(\pi, x) - \inf_{s'_\pi \in \mathcal{H}_\pi} C_{\ell_{\text{def-adv}}}(\pi_{s'_\pi}, x),$$

$$\Delta C_{\Phi^{\text{aug},\tau}}(s_\pi, x) := C_{\Phi^{\text{aug},\tau}}(s_\pi, x) - \inf_{s'_\pi \in \mathcal{H}_\pi} C_{\Phi^{\text{aug},\tau}}(s'_\pi, x).$$

Here $\pi_{s'_\pi}$ denotes the predictor induced by s'_π . When $\|\mathbf{w}(x)\|_1 > 0$, the identities above show that these conditional gaps are exactly the weighted multiclass excess risks induced by $p(x)$, up to the common scale factor $\|\mathbf{w}(x)\|_1$. This is the local reason the augmented surrogate is aligned with the correct comparison. The remainder of the proof turns this pointwise observation into an $\mathcal{H}_\pi$ -consistency statement at the population level.

This identifies the right conditional comparison, but the theorem is an $\mathcal{H}_\pi$ -consistency statement rather than a purely pointwise calibration claim. To keep the hypothesis-class restriction explicit, we gather these conditional weighted problems into the multiclass distribution $\tilde{\mathcal{D}}$ defined in Lemma 19. Under that distribution, the comp-sum loss Φ_{01}^τ satisfies the usual multiclass $\mathcal{H}_\pi$ -consistency inequality with transfer function Γ_τ . Write

$$\begin{aligned}\Delta\mathcal{E}_{01}(\pi; \tilde{\mathcal{D}}) &:= \mathcal{E}_{\ell_{01}}(\pi; \tilde{\mathcal{D}}) - \mathcal{E}_{\ell_{01}}^B(\mathcal{H}_\pi; \tilde{\mathcal{D}}) + \mathcal{U}_{\ell_{01}}(\mathcal{H}_\pi; \tilde{\mathcal{D}}), \\ \Delta\mathcal{E}_{\Phi_{01}^\tau}(s_\pi; \tilde{\mathcal{D}}) &:= \mathcal{E}_{\Phi_{01}^\tau}(s_\pi; \tilde{\mathcal{D}}) - \mathcal{E}_{\Phi_{01}^\tau}^*(\mathcal{H}_\pi; \tilde{\mathcal{D}}) + \mathcal{U}_{\Phi_{01}^\tau}(\mathcal{H}_\pi; \tilde{\mathcal{D}}).\end{aligned}$$

By the symmetry and completeness assumptions on $\mathcal{H}_\pi$ in Theorem 8, the comp-sum $\mathcal{H}_\pi$ -consistency bound of Mao et al. (2023b) applies and gives

$$\Delta\mathcal{E}_{01}(\pi; \tilde{\mathcal{D}}) \leq \Gamma_\tau\left(\Delta\mathcal{E}_{\Phi_{01}^\tau}(s_\pi; \tilde{\mathcal{D}})\right).$$

Lemma 19 shows that this ordinary multiclass problem is exactly the population counterpart of the conditional reduction above. Define

$$\tilde{\Gamma}_\tau(u) := \mathbb{E}[\|\mathbf{w}(X)\|_1] \Gamma_\tau\left(\frac{u}{\mathbb{E}[\|\mathbf{w}(X)\|_1]}\right),$$

so that $\mathbb{E}[\|\mathbf{w}(X)\|_1] \Gamma_\tau(u) = \tilde{\Gamma}_\tau(\mathbb{E}[\|\mathbf{w}(X)\|_1] u)$ by construction. Multiplying the previous inequality by $\mathbb{E}[\|\mathbf{w}(X)\|_1]$ gives

$$\mathbb{E}[\|\mathbf{w}(X)\|_1] \Delta\mathcal{E}_{01}(\pi; \tilde{\mathcal{D}}) \leq \tilde{\Gamma}_\tau\left(\mathbb{E}[\|\mathbf{w}(X)\|_1] \Delta\mathcal{E}_{\Phi_{01}^\tau}(s_\pi; \tilde{\mathcal{D}})\right).$$

We now translate this inequality back to deferral with advice using (25)–(30). The additive term $\mathbb{E}[D(X, A, Y)]$ appears in both the incurred true risk and the best-in-class true risk, so it cancels from the excess-risk difference. We obtain

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\pi) - \mathcal{E}_{\ell_{\text{def-adv}}}^B(\mathcal{H}_\pi) + \mathcal{U}_{\ell_{\text{def-adv}}}(\mathcal{H}_\pi) \leq \tilde{\Gamma}_\tau(\mathcal{E}_{\Phi^{\text{aug}, \tau}}(s_\pi) - \mathcal{E}_{\Phi^{\text{aug}, \tau}}^*(\mathcal{H}_\pi) + \mathcal{U}_{\Phi^{\text{aug}, \tau}}(\mathcal{H}_\pi)),$$

which is exactly (13).

This proves the claimed $\mathcal{H}_\pi$ -consistency bound. When the minimizability gaps vanish, the right-hand side tends to zero whenever the surrogate excess risk tends to zero, so the augmented surrogate recovers the best-in-class deferral-advice risk. If the class is Bayes-rich, this best-in-class risk is the Bayes-optimal deferral-advice risk.

From Mao et al. (2023b), we have:

$$\Gamma_\tau(u) = \begin{cases} \sqrt{2^\tau(2-\tau)u} & \tau \in [0, 1) \\ \sqrt{2|\Pi|^{\tau-1}u} & \tau \in [1, 2) \\ (\tau-1)|\Pi|^{\tau-1}u & \tau \in [2, +\infty) \end{cases} \quad (31)$$

1140

■

1141 B.9 Proof of Corollary 9

Proof The corollary is the statistical form of Theorem 8. Once the two minimizability gaps vanish, the theorem compares the true best-in-class excess risk of the induced policy directly to the surrogate excess risk of the learned score; the Bayes-richness assumption identifies this best-in-class risk with the unrestricted Bayes risk.

Under the stated assumption,

$$\mathcal{U}_{\ell_{\text{def-adv}}}(\mathcal{H}_\pi) = \mathcal{U}_{\Phi^{\text{aug}, \tau}}(\mathcal{H}_\pi) = 0.$$

Moreover, by Bayes-richness,

$$\mathcal{E}_{\ell_{\text{def-adv}}}^B(\mathcal{H}_\pi) = \inf_{\pi} \mathcal{E}_{\ell_{\text{def-adv}}}(\pi).$$

Applying Theorem 8 to the learned score $\widehat{s}_{\pi,n}$ and its induced predictor $\widehat{\pi}_n$ therefore gives, for every n ,

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\widehat{\pi}_n) - \inf_{\pi} \mathcal{E}_{\ell_{\text{def-adv}}}(\pi) \leq \widetilde{\Gamma}_\tau(\mathcal{E}_{\Phi^{\text{aug},\tau}}(\widehat{s}_{\pi,n}) - \mathcal{E}_{\Phi^{\text{aug},\tau}}^*(\mathcal{H}_\pi)).$$

Thus the only remaining quantity to control is the surrogate excess risk.

By assumption,

$$\mathcal{E}_{\Phi^{\text{aug},\tau}}(\widehat{s}_{\pi,n}) - \mathcal{E}_{\Phi^{\text{aug},\tau}}^*(\mathcal{H}_\pi) \rightarrow 0$$

in probability. Since Γ_τ is concave, non-negative, non-decreasing, and satisfies $\Gamma_\tau(0) = 0$, the definition of $\widetilde{\Gamma}_\tau$ in Theorem 8 implies that $\widetilde{\Gamma}_\tau$ is continuous at 0. It therefore follows that

$$\widetilde{\Gamma}_\tau(\mathcal{E}_{\Phi^{\text{aug},\tau}}(\widehat{s}_{\pi,n}) - \mathcal{E}_{\Phi^{\text{aug},\tau}}^*(\mathcal{H}_\pi)) \rightarrow 0$$

in probability as well. Combining this with the preceding excess-risk bound proves that

$$\mathcal{E}_{\ell_{\text{def-adv}}}(\widehat{\pi}_n) - \inf_{\pi} \mathcal{E}_{\ell_{\text{def-adv}}}(\pi) \rightarrow 0$$

in probability. ■

C Experiments

C.1 Global Experimental Protocol

Before turning to the individual tasks, we summarize the experimental choices that are shared across the appendix. The goal is to evaluate the learned policy in the regime studied by the theory. The learner does not predict an abstract label directly. It chooses an executed expert-advice pair, and the value of that choice is revealed only through the resulting realized cost. The experimental protocol is therefore designed to mirror the formal setup as closely as possible.

Evaluation metric. Our primary metric is always the empirical average of the true deferral-advice loss,

$$\widehat{\mathcal{E}}_{\ell_{\text{def-adv}}}(\pi) = \frac{1}{n_{\text{val}}} \sum_{i=1}^{n_{\text{val}}} \ell_{\text{def-adv}}(\pi; x_i, a_i, y_i, \mathbf{e}_i),$$

evaluated on the validation split of the corresponding experiment. This is the only quantity used to compare policies across values of λ . The reason is structural. Once advice costs vary, two policies may have very different consultation patterns even when they achieve similar predictive accuracy. Accuracy alone therefore no longer represents the deployment objective of the paper. The true loss remains the correct comparison criterion because it accounts jointly for task error, expert fees, and advice costs. The appendix therefore keeps the focus on true-loss tables and uses advice-distribution figures only as diagnostics to explain how a method achieves a certain true loss rather than to rank methods.

Learned policy. In every experiment, the learned policy acts on the composite action space $\Pi = [J] \times [K]_0$. The deployed decision is therefore a single executed pair (j, k) . Still, it is useful to parameterize the scores in a way that mirrors the protocol. We write the score of action (j, k) as

$$s_\theta(x, (j, k)) = u_\theta(x, j) + v_\theta(x, j, k),$$

where $u_\theta(x, j)$ plays the role of an expert-routing score and $v_\theta(x, j, k)$ is an expert-conditional advice adjustment. Both depend on the input through a shared learned representation, so the same construction can be used for text and tabular inputs.

A convenient instantiation first maps the input to a latent representation $z_\theta(x) \in \mathbb{R}^d$ and then introduces routing embeddings $a_j \in \mathbb{R}^d$, advice-side expert embeddings $m_j \in \mathbb{R}^d$, advice embeddings $g_k \in \mathbb{R}^d$, and score biases $\rho_j, \delta_k \in \mathbb{R}$:

$$\begin{aligned} u_\theta(x, j) &:= \rho_j + \langle a_j, z_\theta(x) \rangle, \\ v_\theta(x, j, k) &:= \delta_k + \langle m_j, g_k \odot z_\theta(x) \rangle, \end{aligned}$$

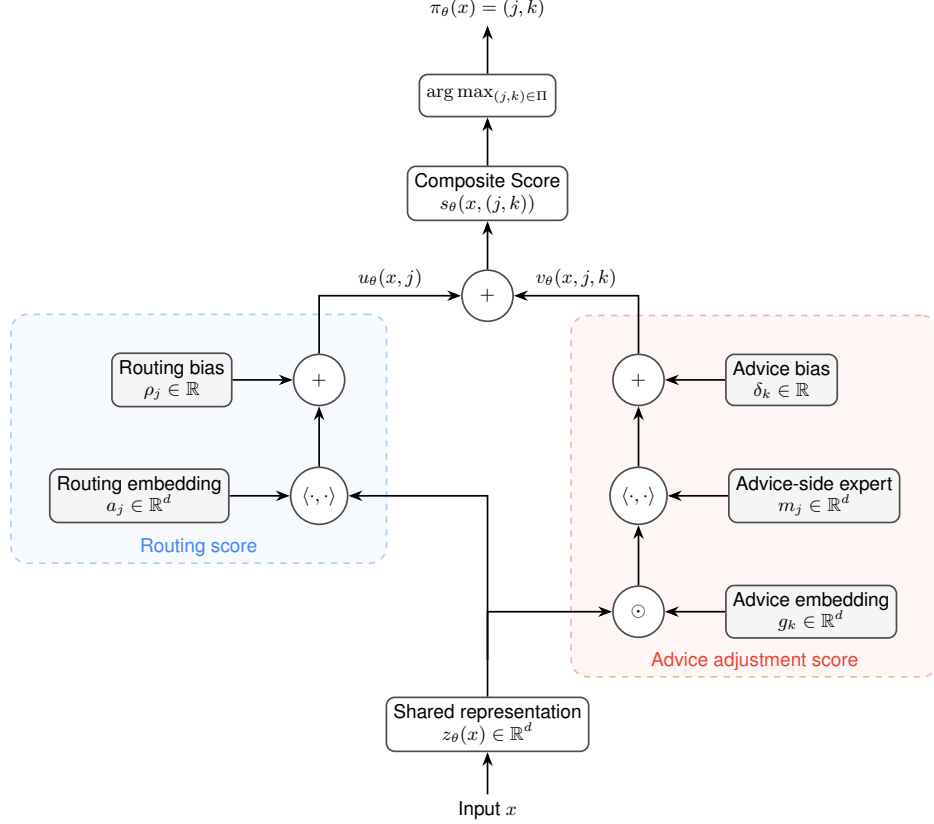


Figure 4: Structured parameterization of the composite policy score. The final decision remains a single argmax over executed expert-advice pairs, while the score decomposes into a routing term and an expert-conditional advice adjustment.

where $\odot$ denotes elementwise multiplication. This factorization preserves three properties that are important in practice. First, routing remains input-dependent through u_θ . Second, the value of advice acquisition can depend jointly on the input and an expert through v_θ . Third, the expert-advice interaction is non-separable, while the number of parameters still scales through a shared low-rank representation rather than independently over all $J(K + 1)$ composite actions.

Decoding remains a single argmax over composite actions:

$$\pi_\theta(x) = \arg \max_{(j,k) \in \Pi} s_\theta(x, (j, k)).$$

Resources. All experiments were conducted on an internal cluster using an NVIDIA A100 GPU with 40 GB of VRAM.

Reading the appendix tables. The main paper reports only the true deferral-advice loss, because that is the deployment metric of interest and the one that remains meaningful across cost regimes. The appendix keeps the same true-loss tables, but adds expert-by-advice summaries and advice-distribution figures for selected values of λ . These additional diagnostics help explain how the learned policy adapts as advice becomes more or less expensive, and they make explicit how the deployment-optimal expert-advice pair changes with the cost regime.

C.1.1 Baselines

We use the same baseline family throughout, and each baseline is included for a specific reason. All methods are evaluated on exactly the same precomputed executed-pair table, so differences in performance come from the decision rule rather than from stochastic variation in the experts or advice sources.

Our method. This is our main method. It predicts directly over the composite action space $\Pi = [J] \times [K]_0$ and is the only learned policy that is allowed to coordinate expert selection and

advice acquisition jointly. Its role is therefore to measure the benefit of solving the full deferral-advice problem.

L2D. This is the standard no-query baseline obtained by restricting the action space to the J actions $(j, 0)$, as established in (Mao et al., 2023a, 2024g; Montreuil et al., 2025b). It answers the first natural question: is explicit advice acquisition useful at all beyond ordinary learning-to-defer? If our method does not improve over L2D, then the added advice mechanism is not buying anything under the true deployment objective.

Best fixed pair. This baseline selects a single constant composite action (j, k) and applies it to every input. It is an oracle over non-adaptive policies, and therefore measures how much is gained by input-dependent decision-making rather than by choosing one good expert-advice pair globally.

Partial-randomization ablations. These are the two most informative diagnostic baselines. In “learned expert, random advice,” we keep the expert chosen by our learned policy and replace only the advice action by a uniform random choice. In “random expert, learned advice,” we do the reverse. Taken together, these two baselines separate the contribution of expert selection from that of advice selection. If the full method improves over both, then the gain cannot be attributed to only one half of the decision.

Random (j, k) . The final baseline samples both the expert and the advice action uniformly. Its role is not diagnostic but calibrating: it provides an uninformed floor against which the learned and structured methods can be compared.

C.2 Synthetic Theorem-Aligned Experiment

This appendix includes one synthetic benchmark whose role is purely theoretical. The goal is not to mimic a realistic deployment pipeline, but to instantiate the exact mechanism of Theorem 6 and show that the observed failure of the separated surrogate is the one predicted by the theory. The same benchmark also contains a second region in which advice is genuinely useful, so that standard L2D is structurally suboptimal. The synthetic experiment therefore isolates the two distinct reasons why the augmented formulation should dominate the alternatives.

Goal, data, and metric. We work in the minimal binary setting of the theorem: two experts ($J = 2$) and one advice action ($K = 1$), so the composite action space is $\Pi = \{(1, 0), (1, 1), (2, 0), (2, 1)\}$. Inputs are sampled uniformly from $[-1, 1]^2$, and the first coordinate partitions the space into two regions:

$$R_- := \{x \in [-1, 1]^2 : x_1 < 0\}, \quad R_+ := \{x \in [-1, 1]^2 : x_1 \geq 0\}.$$

The second coordinate is irrelevant to the Bayes rule; it is included only so that the learned decision maps can be visualized on a two-dimensional grid. We evaluate every policy by its expected true deferral-advice loss on an independent test split drawn from the same synthetic environment. Because the environment is defined directly through executed costs, the evaluation metric matches the object studied in the theory exactly.

Certified cost construction. We set the consultation costs to zero, $\beta_1 = \beta_2 = 0$, and use the advice fees $\gamma_0 = 0$ and $\gamma_1 = 0.08$. The displayed matrices below are conditional expected executed-cost tables,

$$T_{j,k}^\pm = \mathbb{E}[c_{j,k}(X, A, Y) \mid X \in R_\pm], \quad j \in \{1, 2\}, k \in \{0, 1\}.$$

On R_- , we use the certified theorem table

$$T^- = \begin{pmatrix} 0.38 & 1.08 \\ 0.50 & 0.51 \end{pmatrix},$$

while on R_+ we use the advice-helpful table

$$T^+ = \begin{pmatrix} 0.55 & 0.18 \\ 0.30 & 0.90 \end{pmatrix}.$$

Table 3 records these two regions in the same executed-action format as the main paper. The left table is exactly the bad table in Theorem 6 with $b = 0.50$, $\varepsilon = 0.01$, $\delta = 0.12$, and $C = 1.08$. The tables are also realizable under the cost model (3): for each region and pair (j, k) , let the task loss be Bernoulli with mean $p_{j,k}^\pm = T_{j,k}^\pm - \gamma_k$. These probabilities are

$$p^- = \begin{pmatrix} 0.38 & 1.00 \\ 0.50 & 0.43 \end{pmatrix}, \quad p^+ = \begin{pmatrix} 0.55 & 0.10 \\ 0.30 & 0.82 \end{pmatrix},$$

1247 which all lie in $[0, 1]$. Hence $\mathbb{E}[\psi(e_j(X, \tilde{A}^{(k)}), Y) \mid X \in R_{\pm}] + \gamma_k = T_{j,k}^{\pm}$, so the synthetic
 1248 environment is not an abstract cost table detached from the prediction-loss-plus-advice-fee model.

Table 3: Certified conditional executed-cost tables used in the synthetic benchmark. Each row is one region of the input space, and each column is one executed expert-advice pair.

Region	(1, 0)	(1, 1)	(2, 0)	(2, 1)	Bayes action	Bayes risk
R_-	0.380	1.080	0.500	0.510	(1, 0)	0.380
R_+	0.550	0.180	0.300	0.900	(1, 1)	0.180

1249 The left region is the theorem region. Bayes compares row minima and therefore prefers expert 1,
 1250 since $\min_k T_{1,k}^- = 0.38 < 0.50 = \min_k T_{2,k}^-$. The exact binary separated surrogate from the paper
 1251 behaves differently: its profiled row summaries satisfy

$$F(0.38, 1.08) = 0.8371 > 0.7000 = F(0.50, 0.51),$$

1252 so the profiled surrogate reverses the comparison and favors expert 2. The right region serves a
 1253 different purpose. There, Bayes chooses the queried action (1, 1) with cost 0.18, whereas L2D is
 1254 restricted to the no-advice column and must therefore pick expert 2 with cost 0.30. Since both regions
 1255 have positive probability, the separated surrogate and L2D each incur a positive irreducible excess
 1256 risk, but for different reasons.

1257 **Policies and exact separated baseline.** We compare our method, L2D, the same random baselines
 1258 used in the real experiments, and one *exact* implementation of the binary separated surrogate analyzed
 1259 in Section 4.4. This is important. The separated baseline is not an approximate router-query variant.
 1260 It is precisely the special case obtained by setting $\nu = \text{id}$, $G \equiv 1$, $U = \text{id}$, and $\Psi_j = \Phi_{j-1}$, so that
 1261 each indicator in the true binary loss is replaced by its logistic envelope. Concretely, the model uses
 1262 one binary router score s_r^b and one expert-conditional query score $s_{q_1^b}$ or $s_{q_2^b}$ per row, exactly as in
 1263 the theorem statement.

1264 **Training protocol.** All learned methods — our augmented surrogate, the separated baseline, and
 1265 the L2D baseline — share the same backbone: a two-layer MLP over the synthetic input $x \in \mathbb{R}^2$,
 1266 with hidden widths (32, 32) and ReLU activations. They are trained with AdamW (learning rate
 1267 3×10^{-3} , no weight decay, gradient clipping at norm 10) for 120 epochs with batch size 256 and no
 1268 learning-rate scheduler. Our method uses the augmented comp-sum surrogate at $\tau = 1$; the separated
 1269 baseline uses the exact binary logistic surrogate described above; and the L2D baseline is restricted to
 1270 the $k=0$ slice and trained with the standard cross-entropy deferral surrogate. We sweep the training
 1271 size over $n \in \{250, 500, 1000, 2500, 5000\}$ and report averages over five seeds. The figures and
 1272 tables below use the largest train size, $n = 5000$, for the pointwise comparisons.

Table 4: Synthetic test performance at the largest train size ($n = 5000$). Our method is the only one whose excess risk is essentially zero.

Method	Test true risk	Test excess risk	Advice rate (%)	Bayes-action match (%)
Our	0.281 ± 0.001	0.001 ± 0.000	49.6	99.3
Separated	0.334 ± 0.004	0.054 ± 0.004	62.5	58.4
L2D	0.342 ± 0.001	0.062 ± 0.000	0.0	48.8
Learned route, random advice	0.661 ± 0.002	0.381 ± 0.002	49.6	24.5
Random route, no advice	0.432 ± 0.001	0.152 ± 0.001	0.0	24.9
Random (j, k)	0.550 ± 0.003	0.269 ± 0.004	50.1	24.9

1273 Table 4 is the central quantitative result. Our method reaches 0.281 ± 0.001 true risk, essentially
 1274 matching the Bayes optimum of 0.280. The separated surrogate stalls far above this level, with excess
 1275 risk 0.054 ± 0.004 , even though it is trained with the exact binary surrogate analyzed in the theorem.
 1276 L2D also remains suboptimal, but for a different reason: it never queries, so it cannot realize the
 1277 Bayes action on R_+ . The random baselines are much worse, which confirms that the gain is not
 1278 coming from a trivial bias in the environment.

1279 Table 5 shows that the two failure modes are exactly the ones predicted by the theory. On R_- , the
 1280 separated surrogate matches the Bayes action only 18.0% of the time, even though it is nearly perfect
 1281 on R_+ ; this is the empirical footprint of the profiled-row distortion proved in Theorem 6. L2D
 1282 exhibits the opposite pattern. It is nearly optimal on R_- , where the Bayes action is no-advice, but
 1283 it matches the Bayes action 0% of the time on R_+ because the Bayes decision there is the queried

Table 5: Regionwise diagnosis at the largest train size ($n = 5000$). The theorem region R_- isolates the failure of the separated surrogate, while the advice-helpful region R_+ isolates the limitation of L2D.

Method	Risk on R_-	Risk on R_+	Bayes match on R_- (%)	Bayes match on R_+ (%)
Our	0.381	0.181	99.3	99.3
Separated	0.484	0.184	18.0	99.0
L2D	0.383	0.300	97.5	0.0
Learned route, random advice	0.727	0.594	48.9	0.1
Random route, no advice	0.440	0.424	49.8	0.0
Random (j, k)	0.617	0.482	24.7	25.1

1284 action $(1, 1)$. Our method is nearly perfect on both regions, which is exactly the consistency story
 1285 established by Theorem 8.

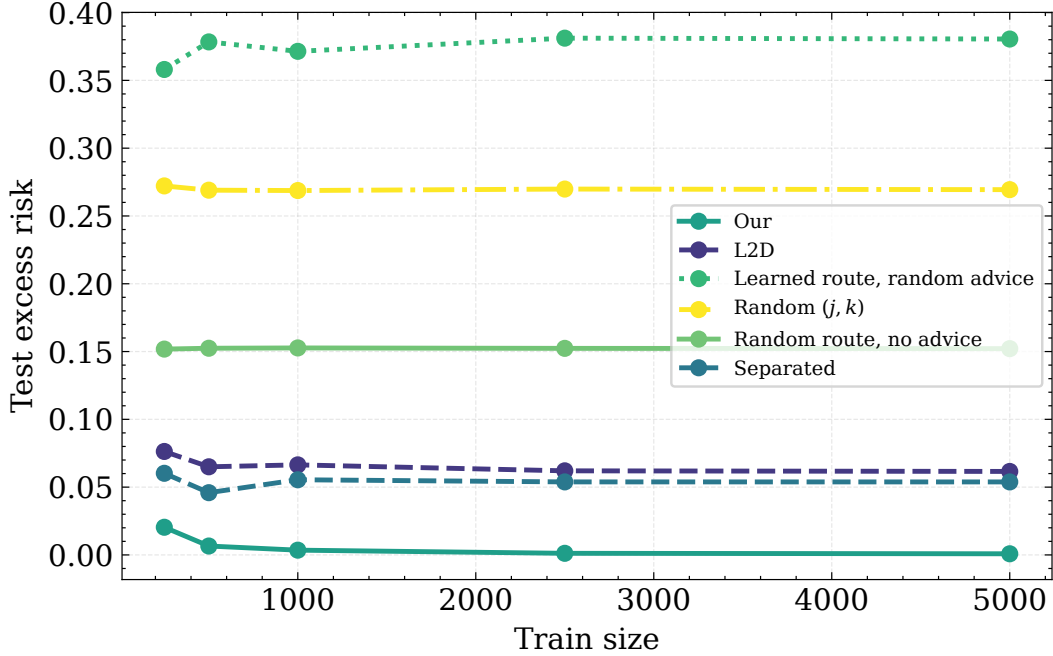


Figure 5: Synthetic test excess risk as a function of train size. Our method approaches zero excess risk, whereas the exact separated surrogate and L2D remain bounded away from the Bayes optimum.

1286 Figures 5 and 6 make the same point visually. The excess risk curve shows that our method is the only
 1287 one that continues to improve toward Bayes optimality as the sample size grows. The action maps
 1288 then show *why*: our method learns the correct composite decision on both halves of the space, while
 1289 the exact separated surrogate and L2D fail on the region singled out by the corresponding theory.

1290 C.3 FEVER Experimental Details

1291 This appendix records the implementation details that are omitted from the main paper for space
 1292 reasons. We focus on the FEVER (Thorne et al., 2018) experiment reported in Section 5, since it is
 1293 the main empirical study in the paper.

1294 **Data, action space, and metric.** FEVER (Thorne et al., 2018) is a fact-verification benchmark built
 1295 from short natural-language claims paired with Wikipedia evidence. Each example must be classified
 1296 as SUPPORTS, REFUTES, or NOT_ENOUGH_INFO depending on whether the claim is supported by the
 1297 evidence available in Wikipedia. A typical example is a claim such as “The Eiffel Tower is located
 1298 in Berlin”; the correct label is REFUTES once the relevant Wikipedia passages are retrieved. The
 1299 experiment uses 15,000 examples sampled from the labelled development split, with a stratified
 1300 75/25 train-validation partition. The composite action space has size 3×5 , corresponding to three

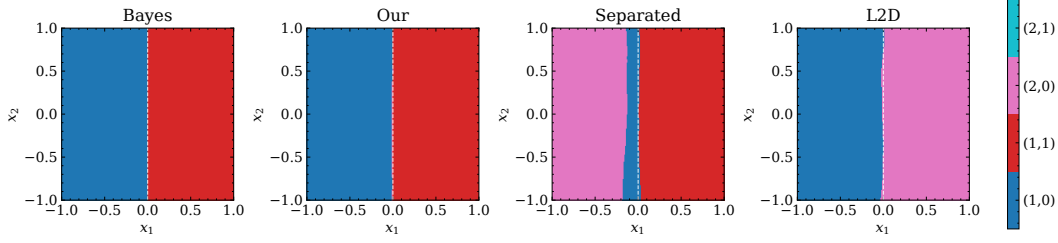


Figure 6: Learned decision maps at the largest train size ($n = 5000$). The vertical dashed line marks the transition between the theorem region R_- and the advice-helpful region R_+ . Our method recovers the Bayes split, the separated surrogate selects the wrong expert on the left region, and L2D cannot realize the queried Bayes action on the right region.

Algorithm 4 FEVER inference example

Require: Claim x , learned composite scorer $s_\pi(\cdot, \cdot; \hat{\theta})$

- 1: Define the FEVER expert set $\{\text{Qwen3-4B-Instruct}, \text{Qwen3-8B}, \text{DeBERTa-v3}\}$
- 2: Define the advice actions $\{0, \text{top-1}, \text{top-3}, \text{top-5}, \text{reformulate+top-5}\}$
- 3: Select the executed pair

$$(\hat{j}, \hat{k}) \leftarrow \arg \max_{(j,k) \in [3] \times [4]_0} s_\pi(x, (j, k); \hat{\theta})$$

- 4: **if** $\hat{k} = 0$ **then**
 - 5: Set $\tilde{a}^{(\hat{k})} \leftarrow \tilde{a}^{(0)}$ and issue no retrieval request
 - 6: **else if** $\hat{k} = \text{top-1}$ **then**
 - 7: Run BM25 on claim x and keep the top-1 retrieved passage
 - 8: Mask all non-selected advice slots to form $\tilde{a}^{(\hat{k})}$
 - 9: **else if** $\hat{k} = \text{top-3}$ **then**
 - 10: Run BM25 on claim x and keep the top-3 retrieved passages
 - 11: Mask all non-selected advice slots to form $\tilde{a}^{(\hat{k})}$
 - 12: **else if** $\hat{k} = \text{top-5}$ **then**
 - 13: Run BM25 on claim x and keep the top-5 retrieved passages
 - 14: Mask all non-selected advice slots to form $\tilde{a}^{(\hat{k})}$
 - 15: **else**
 - 16: Use the reformulation model to rewrite the claim query
 - 17: Mask all non-selected advice slots to form $\tilde{a}^{(\hat{k})}$
 - 18: **end if**
 - 19: Route the claim and masked advice to expert $\hat{j}$
 - 20: Output the FEVER label $e_{\hat{j}}(x, \tilde{a}^{(\hat{k})}) \in \{\text{SUPPORTS}, \text{REFUTES}, \text{NOT ENOUGH INFO}\}$
 - 21: **return** executed pair $\hat{\pi}(x) = (\hat{j}, \hat{k})$ and prediction $e_{\hat{j}}(x, \tilde{a}^{(\hat{k})})$
-

experts and five advice actions. All policy comparisons use the global protocol of Section C.1; in particular, the reported numbers are validation averages of the true deferral-advice loss.

Experts and advice construction. The expert pool contains Qwen3-4B-Instruct (Yang et al., 2025), Qwen3-8B, and DeBERTa-v3-large (He et al., 2021) fine-tuned on entailment tasks, with respective consultation costs $\beta = (0.03, 0.05, 0.04)$. Advice is derived from Wikipedia evidence retrieved with BM25 (Robertson and Zaragoza, 2009). The five advice actions are: no retrieval, BM25 top-1, BM25 top-3, BM25 top-5, and query reformulation followed by BM25 top-5. The reformulated query is produced by Qwen2.5-1.5B. Query costs take the form $\gamma_k = \lambda \gamma_k^{\text{base}}$ with $\gamma^{\text{base}} = (0, 0.015, 0.02, 0.03, 0.01)$. This construction is chosen to induce heterogeneous expert-advice trade-offs, rather than to mimic exact monetary costs.

Prompt templates. The expert-facing prompts are deliberately simple so that the experiment isolates the interaction between routing and advice rather than prompt engineering. For the two Qwen experts, the system instruction is to return exactly one of SUPPORTS, REFUTES, or NOT_ENOUGH_INFO. The user message then instantiates the template

1315 Claim: <claim>.
1316 Evidence: <retrieved passages, or EMPTY when $k = 0$ >.
1317 Answer with exactly one label: SUPPORTS, REFUTES, or
1318 NOT_ENOUGH_INFO.

1319 For example, if the claim is “The Eiffel Tower is located in Berlin,” the top-1 advice may supply a
1320 passage stating that the Eiffel Tower is in Paris, in which case the desired output is REFUTES. The
1321 reformulator is prompted separately to rewrite the claim into short retrieval text, e.g.

1322 Rewrite the claim as a concise Wikipedia search query. Return only
1323 the query text.

1324 The exact prompt strings used in the runs are the ones stored in the released experiment configuration.

1325 **Policy training and baselines.** Input features are frozen all-MiniLM-L6-v2 sentence embeddings
1326 (dimension 384). All learned methods — our augmented surrogate and the L2D baseline — share
1327 the same backbone: a two-layer MLP with hidden dimensions (128, 64), ReLU activations, and no
1328 dropout. They are trained with AdamW (weight decay 10^{-4} , gradient clipping at norm 10) for 50
1329 epochs with batch size 128 and no learning-rate scheduler. We train L2D with a learning rate 5×10^{-3}
1330 and our augmented surrogate with learning rate 10^{-3} . The augmented comp-sum surrogate uses
1331 $\tau = 1$; the L2D baseline uses the same architecture but is restricted to the $k=0$ slice and trained with
1332 the standard cross-entropy deferral surrogate. The best-fixed, partial-randomization, and joint-random
1333 baselines are exactly those defined in Section C.1, instantiated on the FEVER expert pool and advice
1334 set.

Table 6: Validation accuracy (%) of each expert under each advice action at $\lambda = 0$. Bold marks the best advice action for each expert. The preferred action is expert-dependent.

	$k=0$ (none)	$k=1$ (top-1)	$k=2$ (top-3)	$k=3$ (top-5)	$k=4$ (reform.)
Qwen3-4B-Instruct	33.5	33.4	32.9	36.3	34.0
Qwen3-8B	33.5	33.4	34.1	41.1	35.7
DeBERTa-v3-large	33.2	42.9	42.5	42.3	39.2

1335 **Reading the main-paper FEVER table.** Table 6 shows that advice utility is expert-dependent:
1336 the best retrieval strategy is not shared across experts, which is precisely the regime in which a
1337 joint expert-advice policy is needed. Table 1 then evaluates policies on the true deployment metric,
1338 reporting mean and standard deviation over 4 random seeds. Its central message is that our method
1339 achieves the lowest mean loss for seven of the eight retained cost regimes and ties L2D when
1340 advice becomes too expensive at $\lambda = 10$, while also dominating the non-adaptive and randomized
1341 references. The best fixed composite action changes with λ : it is DeBERTa with top-1 retrieval for
1342 $\lambda \in \{0, 0.5, 1, 2, 4\}$, and the cheapest no-advice expert for $\lambda \in \{6, 8, 10\}$.

Table 7: Best advice action for each expert across the retained FEVER cost grid. Each cell reports the best advice index k and the corresponding true loss. Here $k = 0$ denotes no advice, $k = 1$ BM25 top-1, $k = 2$ BM25 top-3, $k = 3$ BM25 top-5, and $k = 4$ reformulation followed by BM25 top-5.

Expert	$\lambda = 0$	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 4$	$\lambda = 6$	$\lambda = 8$	$\lambda = 10$
Qwen3-4B-Instruct	$k=3 / 0.667$	$k=3 / 0.682$	$k=0 / 0.695$	$k=0 / 0.695$	$k=0 / 0.695$	$k=0 / \mathbf{0.695}$	$k=0 / \mathbf{0.695}$	$k=0 / \mathbf{0.695}$
Qwen3-8B	$k=3 / 0.639$	$k=3 / 0.654$	$k=3 / 0.669$	$k=3 / 0.699$	$k=0 / 0.715$	$k=0 / 0.715$	$k=0 / 0.715$	$k=0 / 0.715$
DeBERTa-v3-large	$k=1 / \mathbf{0.611}$	$k=1 / \mathbf{0.619}$	$k=1 / \mathbf{0.626}$	$k=1 / \mathbf{0.641}$	$k=1 / \mathbf{0.671}$	$k=1 / 0.701$	$k=0 / 0.708$	$k=0 / 0.708$

1343 Table 7 makes the mechanism explicit over the full retained grid. At low advice cost, different experts
1344 prefer different advice actions: the two Qwen experts favor deeper retrieval, whereas DeBERTa
1345 prefers the cheapest retrieval action. As λ increases, these choices do not collapse simultaneously.
1346 Qwen3-4B switches to no advice already at $\lambda = 1$, Qwen3-8B remains advice-seeking until $\lambda = 4$,
1347 and DeBERTa keeps top-1 retrieval until $\lambda = 6$ before reverting to no advice at $\lambda = 8$. This staggered
1348 transition is exactly the pattern the joint expert-advice formulation is designed to capture.

1349 Figures 1 and 7 show how our method allocates mass across advice actions. As the advice cost
1350 increases, the policy shifts away from expensive advice, but it does not do so uniformly. The
1351 reallocation toward cheaper actions remains expert-dependent.

1352 Tables 8 and 9 show directly how the deployment-optimal expert-advice pair changes with the cost
1353 regime. At $\lambda = 2$, the lowest loss is achieved by DeBERTa with top-1 retrieval. At $\lambda = 10$, the

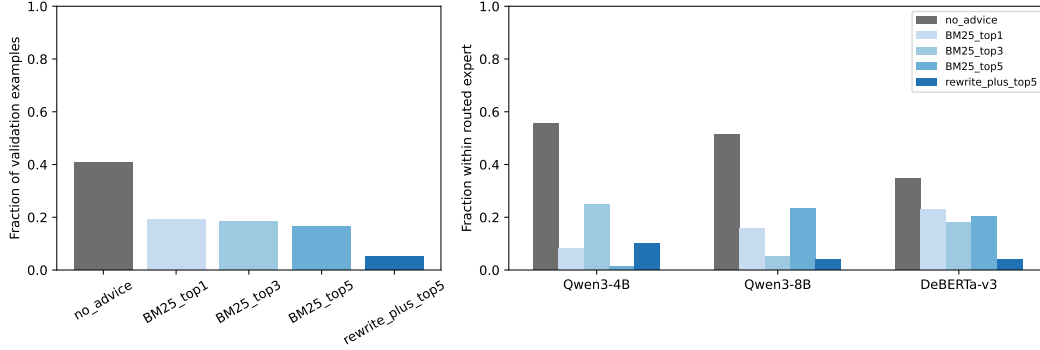


Figure 7: Deployed advice distribution of our method at $\lambda = 2$ in FEVER. Compared with Figure 1, the policy shifts mass away from expensive advice actions, but it does not collapse uniformly to no advice. The expert-conditional panel shows that the transition remains expert-dependent.

Table 8: Expert-by-advice true deferral-advice loss at $\lambda = 2$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
Qwen3-4B-Instruct	0.695	0.726	0.741	0.727	0.710
Qwen3-8B	0.715	0.746	0.749	0.699	0.713
DeBERTa-v3-large	0.708	0.641	0.655	0.677	0.668

optimum shifts to the cheapest no-query expert. What matters operationally is therefore the full deferral-advice loss, not retrieval quality in isolation.

C.4 Sensitive-Escalation Experimental Details

This appendix reports the tabular sensitive-escalation experiment that is deferred from the main paper. The objective is the same as in FEVER, but in a different regime: advice corresponds to progressively more sensitive feature groups rather than retrieved text. The question is whether the composite expert-advice formulation still improves the true deployment objective once the system must trade predictive value against both expert fees and escalation costs.

Data, action space, and metric. We use the IEEE-CIS fraud detection benchmark (Howard et al., 2019). Each example is a payment event with tabular transaction fields, optional identity attributes, and a binary fraud label. A useful operational picture is an online card transaction for which the system initially observes only a restricted low-sensitivity view, such as coarse merchant, amount, and transaction metadata. Before sending the case to an expert, it may decide whether to reveal additional information. For instance, the policy may request contact metadata, a payment-profile summary, a device profile, or stricter identity attributes if these extra signals are worth their escalation cost. We follow the split used in the experiment configuration and train on 120,000 examples with an 8,000-example validation set. With three experts and five advice actions, the composite action space again has size 3×5 . As in FEVER, all policy comparisons follow the global protocol of Section C.1; in particular, the reported numbers are validation averages of the true deferral-advice loss.

Experts and advice construction. The expert pool contains a logistic-regression model, a histogram GBDT, and an MLP, with consultation costs $\beta = (0.02, 0.06, 0.09)$. Advice does not take the form of retrieved text here. Instead, it corresponds to revealing additional feature groups beyond the base low-sensitivity transaction view. The advice levels are nested: each action reveals all information from the previous levels and then adds one more group. Concretely, the five advice actions are:

- $k = 0$: no additional information.
- $k = 1$: the base view plus low-security contact metadata.
- $k = 2$: the information in $k = 1$ plus medium-security payment-profile attributes.
- $k = 3$: the information in $k = 2$ plus high-security device-profile attributes.
- $k = 4$: the information in $k = 3$ plus strict identity attributes.

Table 9: Expert-by-advice true deferral-advice loss at $\lambda = 10$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
Qwen3-4B-Instruct	0.695	0.846	0.901	0.967	0.790
Qwen3-8B	0.715	0.866	0.909	0.939	0.793
DeBERTa-v3-large	0.708	0.761	0.815	0.917	0.748

1383 Their base costs are $\gamma^{\text{base}} = (0, 0.08, 0.12, 0.16, 0.20)$, multiplied by $\lambda \in \{0, 0.04, 0.12, 0.20, 5.0\}$.
1384 Richer advice can improve fraud detection, but each escalation level carries a larger operational and
1385 privacy cost.

1386 The experts themselves are trained offline in a way that matches this nested advice protocol. For
1387 each training example, we do not duplicate the example across all five advice levels. Instead, we
1388 sample one advice level uniformly from $\{0, 1, 2, 3, 4\}$, reveal exactly the corresponding prefix of
1389 feature groups, and train the expert on that masked view augmented with binary reveal indicators.
1390 This produces one memory-safe pooled training set in which each expert sees a mixture of low- and
1391 high-information cases. The logistic model uses balanced class weights, the histogram GBDT uses
1392 class-balanced sample weights, and the MLP additionally oversamples the minority class. After this
1393 offline training stage, each expert is evaluated under all five advice levels on the train and validation
1394 splits to populate the executed expert-advice cost tensor used by the policy learner.

1395 **Policy training and baselines.** Input features are the standardized tabular representation (identity
1396 features with z-score normalization). All learned methods — our augmented surrogate and the L2D
1397 baseline — share the same backbone: a single-hidden-layer MLP with hidden dimension 64, ReLU
1398 activation, and no dropout. They are trained with AdamW (learning rate 10^{-3} , no weight decay,
1399 gradient clipping at norm 10) for 30 epochs with batch size 512 and a cosine learning-rate scheduler
1400 with 5% linear warmup. The augmented comp-sum surrogate uses $\tau = 1$; the L2D baseline uses
1401 the same architecture but is restricted to the $k=0$ slice and trained with the standard cross-entropy
1402 deferral surrogate. For every example and every pair (j, k) , we precompute the expert prediction and
1403 the resulting realized cost, so all methods are trained and evaluated on exactly the same expert-advice
1404 outcome tensor. The best-fixed, partial-randomization, and joint-random baselines are exactly those
1405 defined in Section C.1, instantiated on this fraud-detection expert pool and advice set.

1406 **Reading the appendix sensitive-escalation tables.** The tables below play the same role as their
1407 FEVER counterparts, but in a different modality. Table 10 first asks how each expert behaves when
1408 advice cost changes: which advice action is best for that expert, and how quickly does it revert to no
1409 advice as escalation becomes expensive? Table 11 then evaluates full policies on the true deployment
1410 metric. The detailed tables at $\lambda = 0$, $\lambda = 0.12$, and $\lambda = 5$ make the cost-dependent expert-advice
1411 optimum explicit.

Table 10: Best advice action for each expert across the retained sensitive-escalation cost grid. Each cell reports the best advice index k and the corresponding true loss. Here $k = 0$ denotes no advice, $k = 1$ low-security metadata, $k = 2$ medium-security profile, $k = 3$ high-security device profile, and $k = 4$ strict identity attributes.

Expert	$\lambda = 0$	$\lambda = 0.04$	$\lambda = 0.12$	$\lambda = 0.20$	$\lambda = 5$
Linear	$k=4 / 0.308$	$k=3 / 0.314$	$k=0 / 0.325$	$k=0 / 0.325$	$k=0 / 0.325$
Tree	$k=4 / \mathbf{0.271}$	$k=4 / \mathbf{0.279}$	$k=2 / \mathbf{0.291}$	$k=2 / \mathbf{0.301}$	$k=0 / \mathbf{0.317}$
MLP	$k=2 / 0.359$	$k=2 / 0.364$	$k=0 / 0.369$	$k=0 / 0.369$	$k=0 / 0.369$

1412 Table 10 highlights the same phenomenon as in FEVER in a tabular setting. The best advice action
1413 depends both on the expert and on the cost regime. When escalation is cheap, both the linear model
1414 and the tree expert prefer the most revealing strict-security tier ($k=4$), while the MLP prefers the
1415 medium-security tier ($k=2$). As the escalation cost increases, these preferences shift back toward
1416 cheaper advice or no advice, but not at the same rate for all experts. This is precisely the kind of
1417 expert-dependent trade-off that motivates a joint expert-advice policy.

1418 Table 11 shows a consistent pattern across the entire cost sweep. Our method attains the lowest mean
1419 true loss at every value of λ , with the largest gains over L2D when escalation is cheap and only a
1420 small residual gap at $\lambda = 5$. This is the expected pattern as the optimal policy moves gradually
1421 toward no-advice actions. Our method also stays below the best fixed pair throughout the sweep. That
1422 fixed pair changes with the cost regime, moving from the tree expert with strict-security advice at
1423 $\lambda \in \{0, 0.04\}$, to the same expert with medium-security advice at $\lambda \in \{0.12, 0.20\}$, and finally to the

Table 11: Validation true deferral-advice loss in the sensitive-escalation experiment. Entries are mean $\pm$ standard deviation over 4 random seeds. Lower is better.

Method	$\lambda=0$	$\lambda=0.04$	$\lambda=0.12$	$\lambda=0.20$	$\lambda=5$
Our	0.260 ± 0.001	0.267 ± 0.001	0.274 ± 0.002	0.279 ± 0.002	0.281 ± 0.001
L2D	0.282 ± 0.001	0.282 ± 0.001	0.282 ± 0.001	0.282 ± 0.001	0.282 ± 0.001
Best fixed pair	0.271	0.279	0.291	0.301	0.317
Learned expert, random advice	0.287 ± 0.003	0.297 ± 0.003	0.304 ± 0.003	0.313 ± 0.003	0.837 ± 0.004
Random expert, learned advice	0.310 ± 0.003	0.314 ± 0.003	0.322 ± 0.003	0.325 ± 0.004	0.338 ± 0.003
Random (j, k)	0.325 ± 0.003	0.330 ± 0.003	0.339 ± 0.003	0.348 ± 0.003	0.885 ± 0.006

no-advice tree expert at $\lambda = 5$. The partial-randomization ablations further show that the gain comes from coordinating expert selection and advice acquisition rather than from either component alone; in particular, randomizing the advice becomes especially damaging at high cost, because unnecessary escalation is then heavily penalized.

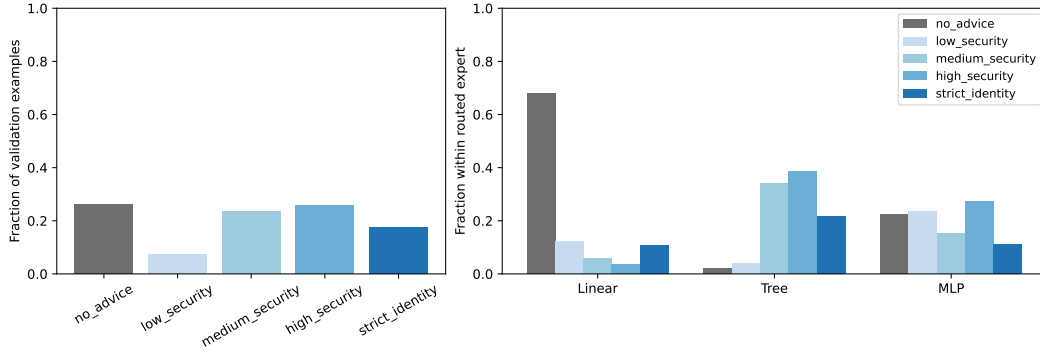


Figure 8: Deployed advice distribution of our method at $\lambda = 0.12$. The left panel shows the overall fraction of validation examples assigned to each advice action. The right panel conditions on the routed expert. At this moderate cost, the learned policy still uses queried advice frequently, but the preferred advice level already depends on the chosen expert.

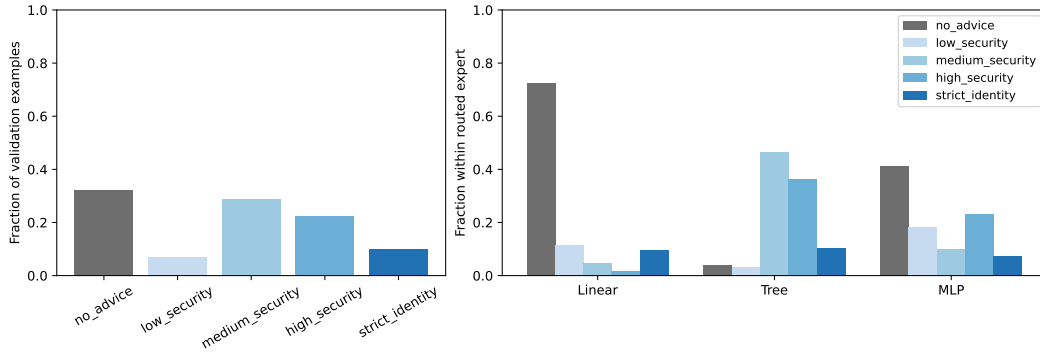


Figure 9: Deployed advice distribution of our method at $\lambda = 0.20$. Compared with Figure 8, the policy shifts further toward cheaper advice and no advice. The right panel shows that this transition is still expert-dependent rather than uniform across the routed experts.

Figures 8 and 9 show how the policy allocates mass across advice levels. The transition with λ is not simply “query” versus “no-query.” As advice becomes more expensive, the learned policy first moves toward cheaper advice levels and only then collapses toward no advice, which explains why the augmented and L2D risks nearly coincide at $\lambda = 5$.

Tables 12, 13, and 14 show how the deployment-optimal action changes with the escalation cost. At $\lambda = 0$, the tree expert with the strict-security advice tier is best. At $\lambda = 0.12$, the optimum has already moved to the medium-security tier for that same expert. At $\lambda = 5$, the best action is the

Table 12: Expert-by-advice true deferral-advice loss at $\lambda = 0$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
Linear	0.325	0.322	0.316	0.308	0.308
Tree	0.317	0.301	0.277	0.273	0.271
MLP	0.369	0.382	0.360	0.377	0.372

Table 13: Expert-by-advice true deferral-advice loss at $\lambda = 0.12$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
Linear	0.325	0.332	0.330	0.327	0.332
Tree	0.317	0.310	0.291	0.292	0.295
MLP	0.369	0.391	0.374	0.396	0.396

no-advice tree expert, and all queried actions are much more costly. The relevant choice is therefore the expert-advice pair with smallest true deferral-advice loss for the current value of λ .

C.5 CLIP Prompt-Escalation Experimental Details

The setting again matches the deferral-advice protocol, but advice now corresponds to increasingly rich prompt families for frozen CLIP (Radford et al., 2021) experts. The question is whether a joint expert-advice policy improves the true deployment objective when richer prompt sets may improve zero-shot accuracy but also incur an explicit consultation cost.

Data, action space, and metric. We use ImageNet-1k (Russakovsky et al., 2015). Each example is an image paired with one of 1000 class labels. A useful way to read this experiment is as a cost-sensitive zero-shot classification problem: given an image of, say, a goldfish, the system may either classify it with a single canonical prompt such as “a photo of a goldfish” or pay for a richer prompt family before sending the image to a CLIP expert. The retained run uses 39,222 training examples and 14,288 validation examples. With three experts and four advice actions, the composite action space has size 3×4 . As in FEVER and sensitive-escalation, all policy comparisons follow the global protocol of Section C.1; in particular, the reported numbers are validation averages of the true deferral-advice loss.

Experts and advice construction. The expert pool contains three frozen CLIP models: CLIP-B/32, CLIP-B/16, and CLIP-L/14, with consultation costs $\beta = (0, 0.04, 0.08)$. Advice does not reveal new image features. Instead, it enlarges the prompt family used to build the text classifier for a given expert. The four advice actions are:

- $k = 0$: no advice, using a single canonical prompt.
- $k = 1$: a clean prompt ensemble with 5 closely related templates.
- $k = 2$: a medium-diversity family with 16 templates.
- $k = 3$: a large-diversity family with 32 templates.

These advice levels are nested in the practical sense that larger prompt families include and extend the simpler ones. Their base costs are $\gamma^{\text{base}} = (0, 0.03, 0.06, 0.09)$, multiplied by $\lambda \in \{0, 0.1, 0.15, 0.3, 0.4, 0.6, 0.7, 0.8, 1.0, 1.5, 10.0\}$. This produces a transparent trade-off: richer prompt families can improve zero-shot accuracy, but only at an additional advice cost.

Prompt examples. It is useful to make these advice actions concrete. For an image whose class is goldfish, the no-advice action uses only the canonical template “a photo of a goldfish.” The clean-ensemble advice augments this with closely related variants such as “a clean photo of a goldfish,” “a centered photo of a goldfish,” and “a high quality photo of a goldfish.” The medium-diversity advice expands further to prompts such as “a close-up photo of a goldfish,” “a bright photo of a goldfish,” “a blurry photo of a goldfish,” and “a black and white photo of a goldfish.” Finally, the large-diversity advice adds still broader views, for instance “a rendering of a goldfish,” “a sketch of a goldfish,” “a nighttime photo of a goldfish,” and “a front-view photo of a goldfish.” In this experiment, acquiring advice therefore means paying to evaluate the image against a richer prompt family before handing it to the selected CLIP expert.

Policy training and baselines. Input features are frozen CLIP-B/32 image embeddings (dimension 512), standardized with z-score normalization. All learned methods — our augmented surrogate and the L2D baseline — share the same backbone: a two-layer MLP with hidden dimensions (512, 256), ReLU activations, and no dropout. They are trained with AdamW (weight decay 10^{-3} ,

Table 14: Expert-by-advice true deferral-advice loss at $\lambda = 5$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$
Linear	0.325	0.722	0.916	1.108	1.308
Tree	0.317	0.701	0.877	1.073	1.271
MLP	0.369	0.782	0.960	1.177	1.372

gradient clipping at norm 10) for 400 epochs with batch size 256 and a cosine learning-rate scheduler with 5% linear warmup. We use a learning rate of 10^{-2} for L2D, and 10^{-3} for our approach. The augmented comp-sum surrogate uses $\tau = 1$; the L2D baseline uses the same architecture but is restricted to the $k=0$ slice and trained with the standard cross-entropy deferral surrogate. For every image, every expert, and every prompt family, we precompute the expert prediction and the realized cost, so all methods are trained and evaluated on exactly the same expert-advice outcome tensor. The best-fixed, partial-randomization, and joint-random baselines are exactly those defined in Section C.1, instantiated on this CLIP expert pool and prompt-family advice set.

Reading the appendix CLIP tables. The tables below mirror those of the previous subsections. Table 15 first shows how each CLIP expert’s preferred advice action changes with the advice cost. Table 16 then evaluates full policies on the true deployment metric, and the detailed tables at $\lambda = 0$, $\lambda = 0.15$, and $\lambda = 10$ make the cost-dependent expert-advice optimum explicit.

Table 15: Best advice action for each expert across the retained CLIP cost grid. Each cell reports the best advice index k and the corresponding true loss. Here $k = 0$ denotes the canonical no-advice prompt, $k = 1$ the clean ensemble, $k = 2$ the medium-diversity prompt family, and $k = 3$ the large-diversity prompt family.

Expert	$\lambda = 0$	$\lambda = 0.1$	$\lambda = 0.15$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.6$	$\lambda = 0.7$	$\lambda = 0.8$	$\lambda = 1$	$\lambda = 1.5$	$\lambda = 10$
CLIP-B/32	$k=3 / 0.391$	$k=2 / 0.398$	$k=2 / 0.401$	$k=0 / 0.403$	$k=0 / 0.403$	$k=0 / 0.403$	$k=0 / 0.403$	$k=0 / 0.403$	$k=0 / 0.403$	$k=0 / 0.403$	$k=0 / 0.403$
CLIP-B/16	$k=3 / 0.378$	$k=2 / 0.386$	$k=2 / 0.389$	$k=0 / 0.394$	$k=0 / 0.394$	$k=0 / 0.394$	$k=0 / 0.394$	$k=0 / 0.394$	$k=0 / 0.394$	$k=0 / 0.394$	$k=0 / 0.394$
CLIP-L/14	$k=3 / 0.351$	$k=1 / 0.359$	$k=1 / 0.361$	$k=1 / 0.365$	$k=0 / 0.366$	$k=0 / 0.366$	$k=0 / 0.366$	$k=0 / 0.366$	$k=0 / 0.366$	$k=0 / 0.366$	$k=0 / 0.366$

Table 15 shows the same structural pattern as the FEVER and tabular experiments. The best advice action depends on both the expert and the cost regime. When advice is cheap, all three experts prefer queried prompt families, although not the same ones: the two smaller CLIP models move through the medium-diversity family, whereas CLIP-L/14 already switches to the cheaper clean ensemble at $\lambda = 0.1$ and $\lambda = 0.15$. Once the advice cost increases further, all three experts revert to the no-advice action, but not at the same point. This is again the regime in which a joint expert-advice policy is most natural.

Table 16: Validation true deferral-advice loss in the CLIP prompt-escalation experiment. Entries are mean $\pm$ standard deviation over 4 random seeds. Lower is better.

λ	Our	L2D	Best fixed pair	Learned expert, random advice	Random expert, learned advice	Random (j, k)
0	0.337 $\pm$ 0.000	0.352 $\pm$ 0.001	0.351	0.343 $\pm$ 0.001	0.374 $\pm$ 0.002	0.379 $\pm$ 0.003
0.1	0.345 $\pm$ 0.000	0.352 $\pm$ 0.001	0.359	0.348 $\pm$ 0.001	0.381 $\pm$ 0.002	0.383 $\pm$ 0.003
0.15	0.348 $\pm$ 0.001	0.352 $\pm$ 0.001	0.361	0.350 $\pm$ 0.001	0.384 $\pm$ 0.002	0.386 $\pm$ 0.003
0.3	0.350 $\pm$ 0.002	0.352 $\pm$ 0.001	0.365	0.357 $\pm$ 0.001	0.387 $\pm$ 0.002	0.392 $\pm$ 0.003
0.4	0.350 $\pm$ 0.002	0.352 $\pm$ 0.001	0.366	0.361 $\pm$ 0.001	0.388 $\pm$ 0.003	0.397 $\pm$ 0.003
0.6	0.350 $\pm$ 0.002	0.352 $\pm$ 0.001	0.366	0.370 $\pm$ 0.001	0.388 $\pm$ 0.002	0.406 $\pm$ 0.003
0.7	0.350 $\pm$ 0.002	0.352 $\pm$ 0.001	0.366	0.374 $\pm$ 0.001	0.388 $\pm$ 0.002	0.410 $\pm$ 0.003
0.8	0.351 $\pm$ 0.002	0.352 $\pm$ 0.001	0.366	0.379 $\pm$ 0.001	0.388 $\pm$ 0.002	0.415 $\pm$ 0.003
1	0.351 $\pm$ 0.001	0.352 $\pm$ 0.001	0.366	0.388 $\pm$ 0.001	0.388 $\pm$ 0.002	0.424 $\pm$ 0.003
1.5	0.351 $\pm$ 0.001	0.352 $\pm$ 0.001	0.366	0.410 $\pm$ 0.001	0.388 $\pm$ 0.002	0.446 $\pm$ 0.003
10	0.352 $\pm$ 0.001	0.352 $\pm$ 0.001	0.366	0.795 $\pm$ 0.003	0.388 $\pm$ 0.002	0.829 $\pm$ 0.004

Table 16 shows a clear cost-dependent transition. Our method attains the lowest mean true loss throughout the low- and moderate-cost regime, improving on both L2D and the best fixed pair. As λ increases, the gap narrows, which is expected because the optimal behavior approaches the no-advice regime. At $\lambda = 10$, our method ties L2D, consistent with the fact that querying has become too expensive to justify and the learned policy ceases to acquire advice. The partial-randomization baselines support the same conclusion as in the other experiments: the gain comes from coordinating expert choice and advice choice rather than from either component in isolation.

Figures 10 and 11 show how the queried mass is distributed across prompt families. As the advice cost increases, the policy moves toward cheaper prompt families and then toward no advice, and it does so differently for different routed experts.

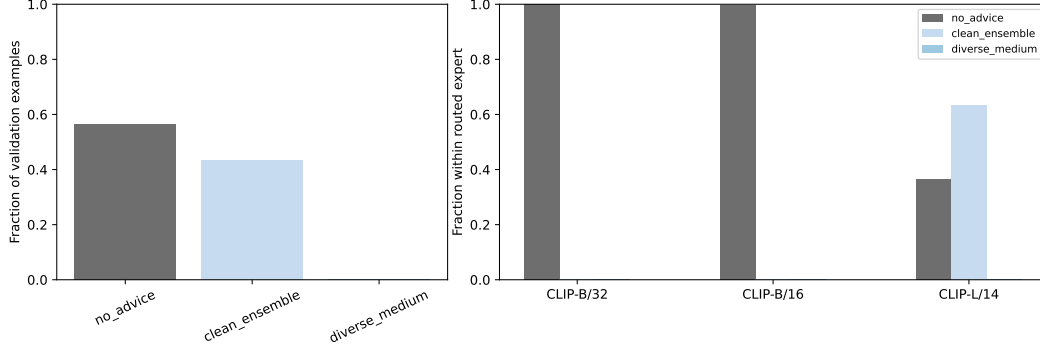


Figure 10: Deployed advice distribution of our method at $\lambda = 0.3$ in the CLIP prompt-escalation experiment. The left panel shows the overall fraction of validation images assigned to each advice action. The right panel conditions on the routed expert. At this cost level, the policy still queries frequently, but it already favors cheaper prompt families than in the free-advice regime.

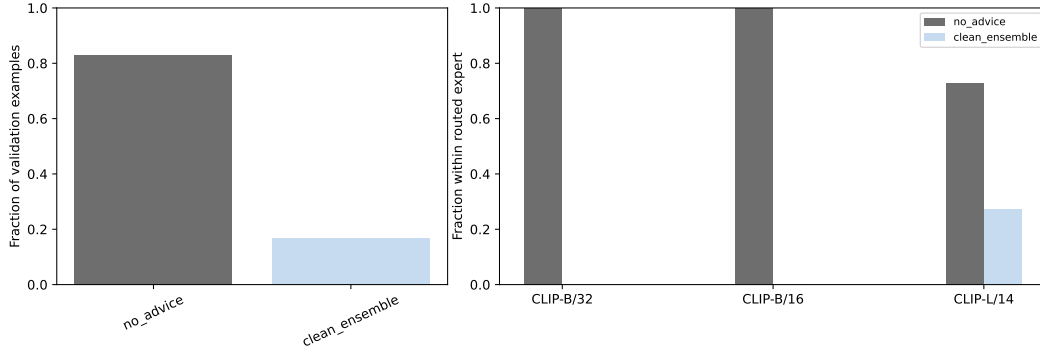


Figure 11: Deployed advice distribution of our method at $\lambda = 0.4$ in the CLIP prompt-escalation experiment. Compared with Figure 10, the policy already shifts further toward no advice and the cheapest prompt families. The expert-conditional panel shows that this transition remains expert-dependent rather than uniform across routed experts.

Table 17: Expert-by-advice true deferral-advice loss at $\lambda = 0$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$
CLIP-B/32	0.403	0.397	0.392	0.391
CLIP-B/16	0.394	0.385	0.380	0.378
CLIP-L/14	0.366	0.356	0.354	0.351

Table 18: Expert-by-advice true deferral-advice loss at $\lambda = 0.15$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$
CLIP-B/32	0.403	0.402	0.401	0.405
CLIP-B/16	0.394	0.390	0.389	0.391
CLIP-L/14	0.366	0.361	0.363	0.364

Table 19: Expert-by-advice true deferral-advice loss at $\lambda = 10$. Lower is better.

	$k=0$	$k=1$	$k=2$	$k=3$
CLIP-B/32	0.403	0.697	0.992	1.291
CLIP-B/16	0.394	0.685	0.980	1.278
CLIP-L/14	0.366	0.656	0.954	1.251

1506 Tables 17, 18, and 19 show how the deployment-optimal prompt family changes with the advice cost.
 1507 At $\lambda = 0$, all three experts prefer the richest prompt family. At $\lambda = 0.15$, the optimum has already

1508 shifted to smaller prompt sets. By $\lambda = 10$, the no-advice action is uniformly optimal in true loss. The
1509 relevant decision is therefore the prompt family with smallest deferral-advice loss, not the richest
1510 family available.

1511 **NeurIPS Paper Checklist**

1512 **1. Claims**

1513 Question: Do the main claims made in the abstract and introduction accurately reflect the
1514 paper’s contributions and scope?

1515 Answer: [\[Yes\]](#)

1516 Justification: The abstract and Section 1 state four contributions (formulation and Bayes-
1517 optimal policy, Bayes inconsistency of separated surrogates, augmented surrogate with
1518 $\mathcal{H}$ -consistency and excess-risk transfer, experiments). Each is supported by a formal result
1519 (Lemma 3, Theorem 6, Theorem 8, Corollary 9) and by experiments in Section 5 and
1520 Appendices C.3–C.2.

1521 **2. Limitations**

1522 Question: Does the paper discuss the limitations of the work performed by the authors?

1523 Answer: [\[Yes\]](#)

1524 Justification: A dedicated Limitations section (Section 7 in the main text) discusses the
1525 full-information training assumption (precomputed cost tensors for every expert–advice
1526 pair) and the resulting limited-information extension left for future work.

1527 **3. Theory assumptions and proofs**

1528 Question: For each theoretical result, does the paper provide the full set of assumptions and
1529 a complete (and correct) proof?

1530 Answer: [\[Yes\]](#)

1531 Justification: All lemmas, theorems, and corollaries are numbered and cross-referenced.
1532 Full proofs with explicit regularity and hypothesis-set conditions are given in Appendix B.
1533 The main text states the assumptions for the central results and gives proof intuition for the
1534 main negative and positive surrogate results.

1535 **4. Experimental result reproducibility**

1536 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
1537 perimental results of the paper to the extent that it affects the main claims and/or conclusions
1538 of the paper (regardless of whether the code and data are provided or not)?

1539 Answer: [\[Yes\]](#)

1540 Justification: Section 5 describes the task, expert pool, advice actions, cost structure, and
1541 baselines. Appendix C.1 describes the shared evaluation protocol and policy parameteriza-
1542 tion; Appendices C.3, C.4, C.5, and C.2 give per-benchmark details.

1543 **5. Open access to data and code**

1544 Question: Does the paper provide open access to the data and code, with sufficient instruc-
1545 tions to faithfully reproduce the main experimental results, as described in supplemental
1546 material?

1547 Answer: [\[Yes\]](#)

1548 Justification: Code to reproduce all experiments is provided in the supplementary material,
1549 together with configuration files and scripts for each benchmark. All datasets used (FEVER,
1550 IEEE-CIS fraud, ImageNet-1k) are publicly available, and the synthetic benchmark is fully
1551 specified by the cost tables in Appendix C.2.

1552 **6. Experimental setting/details**

1553 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
1554 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1555 Answer: [\[Yes\]](#)

1556 Justification: Section 5 summarizes the core experimental setup. Appendix C.1 gives the
1557 shared protocol, and Appendices C.3, C.4, C.5, and C.2 give the per-benchmark architectures,
1558 optimizers, learning rates, seeds, and data splits.

1559 **7. Experiment statistical significance**

1560 Question: Does the paper report error bars suitably and correctly defined or other appropriate
1561 information about the statistical significance of the experiments?

1562 Answer: [Yes]

1563 Justification: Table 1 and the appendix tables report mean $\pm$ standard deviation over 4
1564 random seeds for all learned methods (the synthetic experiment uses 5 seeds). The table
1565 captions state the number of seeds and that reported intervals are standard deviations.

1566 **8. Experiments compute resources**

1567 Question: For each experiment, does the paper provide sufficient information on the com-
1568 puter resources (type of compute workers, memory, time of execution) needed to reproduce
1569 the experiments?

1570 Answer: [Yes]

1571 Justification: Appendix C.1 reports the compute hardware used for the experiments: an
1572 NVIDIA A100 GPU with 40 GB of VRAM.

1573 **9. Code of ethics**

1574 Question: Does the research conducted in the paper conform, in every respect, with the
1575 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

1576 Answer: [Yes]

1577 Justification: The research uses publicly available datasets and open-source pretrained
1578 models. No human subjects are involved. The work conforms with the NeurIPS Code of
1579 Ethics.

1580 **10. Broader impacts**

1581 Question: Does the paper discuss both potential positive societal impacts and negative
1582 societal impacts of the work performed?

1583 Answer: [Yes]

1584 Justification: A dedicated Impact Statement (Section 8) discusses potential benefits (cost-
1585 aware expert allocation, information-aware routing, more transparent automation/human-
1586 reviewer trade-offs) and risks (inheriting expert and cost-structure biases; privacy/fairness
1587 implications of acquiring sensitive advice), and recommends application-specific audits
1588 before deployment.

1589 **11. Safeguards**

1590 Question: Does the paper describe safeguards that have been put in place for responsible
1591 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
1592 image generators, or scraped datasets)?

1593 Answer: [N/A]

1594 Justification: The paper does not release pretrained models, datasets, or other assets that
1595 pose a risk for misuse.

1596 **12. Licenses for existing assets**

1597 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
1598 the paper, properly credited and are the license and terms of use explicitly mentioned and
1599 properly respected?

1600 Answer: [Yes]

1601 Justification: All datasets and models used are cited (FEVER, IEEE-CIS fraud detection,
1602 ImageNet-1k, Qwen3, DeBERTa-v3, CLIP), and no external dataset or model is redistributed
1603 by the paper.

1604 **13. New assets**

1605 Question: Are new assets introduced in the paper well documented and is the documentation
1606 provided alongside the assets?

1607 Answer: [Yes]

1608 Justification: The supplementary material provides the code, configuration files, and scripts
1609 for the experiments. The paper does not introduce a new dataset or pretrained model; the
1610 synthetic benchmark is specified in Appendix C.2.

1611 **14. Crowdsourcing and research with human subjects**

1612 Question: For crowdsourcing experiments and research with human subjects, does the paper

1613 include the full text of instructions given to participants and screenshots, if applicable, as

1614 well as details about compensation (if any)?

1615 Answer: [N/A]

1616 Justification: The paper does not involve crowdsourcing or research with human subjects.

1617 **15. Institutional review board (IRB) approvals or equivalent for research with human**

1618 **subjects**

1619 Question: Does the paper describe potential risks incurred by study participants, whether

1620 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

1621 approvals (or an equivalent approval/review based on the requirements of your country or

1622 institution) were obtained?

1623 Answer: [N/A]

1624 Justification: The paper does not involve human subjects research.

1625 **16. Declaration of LLM usage**

1626 Question: Does the paper describe the usage of LLMs if it is an important, original, or

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1628 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

1629 scientific rigor, or originality of the research, declaration is not required.

1630 Answer: [N/A]

1631 Justification: Any LLM assistance in preparing the manuscript was limited to language

1632 polishing and did not affect the methodology, theoretical results, experiments, or scientific

1633 analysis.